

QuickHire

On-Demand Short-Term Labour Hiring Platform

Final Year Project Documentation

Submitted By

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1. Introduction

Pakistan has a workforce of over 70 million people, a large portion of whom are daily-wage workers, domestic helpers, skilled tradespeople, and semi-skilled labourers. Despite this vast and available workforce, there is no structured, trusted, or digital platform to connect these workers with clients who need them on flexible, short-term terms.

Families needing a cook for Eid week, a cleaner before a wedding, a nurse for ten days post-surgery, or a caretaker for a newborn have no digital solution. Event organizers and businesses seeking temporary staff equally rely on informal WhatsApp groups, word-of-mouth referrals, and local agents — with zero accountability, no verified profiles, and no digital record of past work.

QuickHire is a web-based two-sided marketplace that addresses this gap directly. It connects individuals, families, and businesses with verified workers for short-term, flexible engagements — from a single hour to a multi-week assignment. The platform is designed specifically for Pakistan's informal labour market and is inspired by the model of Urban Company in India, which transformed how people hire home service workers in a comparable socioeconomic context.

A practical example illustrates the platform's value: A family in Lahore welcomes a newborn. They need a certified caretaker for three weeks. They visit QuickHire, filter by category (Childcare), location (Lahore), and duration (three weeks). They browse verified worker profiles, read ratings from previous clients, and confirm a booking. Both parties receive confirmation. The worker arrives. After the period ends, the family rates the worker and the worker receives a verified review on her profile. This entire journey — which today happens through WhatsApp groups and local agents — is replaced by a clean, safe, and accountable digital platform.

1.1 Problem Statement

Pakistan's informal labour market suffers from three critical, interconnected gaps that QuickHire is designed to solve:

Gap 1 — The Duration Gap: Every existing Pakistani hiring platform is built for permanent or long-term employment. No platform treats short-term, per-day, or per-event hiring as the primary use case. Millions of Pakistani households require domestic workers, nurses, cooks, and event staff for durations ranging from a single day to a few weeks, with no digital solution to support this need.

Gap 2 — The Trust Gap: Hiring through WhatsApp groups or a neighbour's recommendation provides zero accountability. There are no verified identities, no public ratings, no reviews, and no digital record of a worker's history. This is especially critical when families are hiring workers for childcare, nursing, and domestic work inside their homes.

Gap 3 — The Worker Dignity Gap: Most existing platforms serve only the client side. Workers have no digital identity, no professional profile, and no way to build or demonstrate a verifiable reputation. A caretaker with five years of experience has no mechanism to prove it or build a career digitally.

1.2 Objectives

The primary objectives of this project are:

- To develop a web-based two-sided marketplace that enables clients to post short-term job listings and workers to apply and be booked for those engagements.
- To implement a worker verification system through national ID upload and admin approval, establishing a trusted and accountable workforce directory.
- To design and build a two-way rating and review system, creating mutual accountability between clients and workers after each completed booking.

- To integrate an AI-powered Smart Worker Recommendation Engine that ranks workers by suitability based on category, location, rating, availability, and experience.
- To provide workers with a verified digital profile and career identity, enabling them to accumulate a job history, ratings, and professional reputation on the platform.
- To architect the system with a REST API layer that supports the future development of a mobile application without requiring a backend rewrite.

1.3 Scope of the Project

Included in Scope:

- Development of a fully responsive, web-based platform with three distinct user roles: Client, Worker, and Admin.
- Client functionality: job posting with category, city, duration, and pay rate; worker browsing and booking; rating and review after job completion.
- Worker functionality: registration and national ID upload; profile building with skills, availability calendar, and job history; job browsing and application.
- Admin dashboard: user management, worker ID verification approvals, platform analytics, and dispute handling.
- Booking lifecycle management: Pending → Confirmed → Completed states with automated email notifications at each transition.
- AI features: Smart Worker Recommendation Engine, Review Sentiment Analysis, and Smart Pay Rate Suggestion.
- Support for 14 service categories including domestic help, childcare, elder care, event staffing, cooking, driving, construction, security, gardening, tutoring, beauty, handyman, moving, and office support.

Out of Scope (Version 1.0):

- Online payment processing — all payments are handled as cash or direct bank transfer between parties.
- SMS notifications — communication is managed through email and in-app notifications.
- GPS-based or real-time location tracking — worker location is handled via city and area text input.
- Native mobile application — Version 1.0 is a mobile-responsive web application only.
- Automated dispute resolution — disputes are handled manually by the platform administrator.

1.4 Significance of the Project

QuickHire addresses a structural gap in Pakistan's labour market that affects tens of millions of people on both the client and worker sides. The significance of this project extends across economic, social, and technological dimensions.

Economically, the platform reduces the cost and friction of short-term hiring for households, event organizers, and businesses, while simultaneously giving informal workers a structured channel to find consistent, paid work. By formalizing a segment of the economy that currently operates entirely through personal networks and agents, QuickHire has the potential to increase income stability for daily-wage workers and reduce hiring costs for clients.

Socially, the platform introduces the trust layer that Pakistan's informal labour sector currently lacks — through identity verification, public ratings, and booking history. This is particularly

significant for families hiring workers into their homes for childcare and nursing, where accountability is a safety concern, not merely a convenience.

From a technological standpoint, QuickHire is the first platform in Pakistan to treat short-term labour hiring as a first-class digital use case, with a full-stack architecture designed for scalability, a REST API layer for future mobile extension, and integrated LLM-powered AI features for intelligent matching, search, and support.

1.5 Artificial Intelligence Features

QuickHire integrates six AI-powered features, built primarily around large language models (LLMs) and retrieval-augmented generation (RAG), to enhance the quality of matches, the reliability of reviews, the efficiency of job posting, the responsiveness of support, and the integrity of the platform. These features are implemented as a dedicated AI service layer that the Django backend calls via internal API requests, using an LLM provider (e.g., Groq, OpenAI, or Anthropic) alongside vector embeddings and a vector database for semantic search.

1. LLM-Powered Job Description Parser (Structured Output Extraction)

When a client writes a free-text job description, the system sends it to an LLM together with a structured-output schema that extracts job category, required skills, location, and expected duration. Unlike traditional rule-based NLP parsers, the LLM can correctly interpret varied phrasing, spelling variation, and implicit detail without hand-engineered extraction rules. This parsed data directly feeds into the worker-matching pipeline, improving match quality without requiring additional form inputs from the client.

This feature allows the platform to:

- Reduce client effort and friction during the job posting process.
- Improve the accuracy and consistency of structured job data compared to manual entry or rule-based parsing.
- Increase job completion rates by ensuring accurate, structured job requirements from the start.

2. AI-Powered Natural-Language Worker Search (Retrieval-Augmented Matching)

Clients can describe their hiring need in plain language (for example, “need someone for elderly care, two weeks, Lahore”). The system converts this query into a vector embedding and performs a semantic similarity search against embedded worker profiles stored in a vector database (e.g., pgvector, Chroma, or Pinecone), retrieving relevant candidates even when exact keywords do not match. Retrieved candidates are then re-ranked using a weighted suitability score:

$$\text{Score} = (\text{Category Match} \times 0.35) + (\text{Location} \times 0.25) + (\text{Rating} \times 0.20) + (\text{Availability} \times 0.10) + (\text{Experience} \times 0.10)$$

This hybrid retrieval-plus-ranking approach combines the flexibility of semantic search with the interpretability of a transparent scoring formula.

This feature allows the platform to:

- Allow clients to search naturally instead of relying only on fixed category and location filters.
- Deliver personalised, real-time worker suggestions immediately after a job is posted.
- Improve worker utilisation rates by matching them with the most relevant job opportunities.

3. LLM-Based Review Sentiment and Theme Analysis

After a job is completed, written reviews from clients are sent to an LLM, which classifies each review as Positive, Neutral, or Negative, extracts key descriptive themes (such as ‘punctual’,

‘reliable’, or ‘rude’), and flags reviews that appear potentially fake or inconsistent, along with a brief natural-language justification for the flag. This contextual, reasoning-based approach captures nuance — such as sarcasm or mixed sentiment — that simple keyword- or lexicon-based sentiment tools often miss.

This feature enables the platform to:

- Provide immediate, accurate feedback summaries for workers in a readable keyword format.
- Help clients make informed hiring decisions based on authentic, context-aware sentiment data.
- Flag potentially fake or misleading reviews for manual moderation by the admin, along with the model's reasoning.

4. AI Support and Onboarding Assistant (Conversational RAG)

A chat-based assistant answers worker and client questions about how the platform works — such as booking steps, verification requirements, and payment handling — by retrieving relevant sections from QuickHire's FAQ and policy documents and generating grounded, conversational answers using an LLM. This reduces reliance on manual admin support for repetitive queries.

This feature enables the platform to:

- Provide instant, always-available answers to common onboarding and platform questions.
- Reduce the support burden on platform administrators for repetitive queries.
- Improve the first-time experience for new clients and workers unfamiliar with the platform.

5. AI Dispute-Summary Assistant (Admin-Facing LLM Reasoning)

When a client and worker raise a dispute, the system feeds the relevant booking details and message history into an LLM, which generates a structured summary of the disagreement along with a suggested resolution for the administrator to review. This does not automate dispute resolution but accelerates the administrator's review process by surfacing the key facts and a starting recommendation.

This feature enables the platform to:

- Reduce the time administrators spend reading through full conversation histories.
- Provide a consistent, structured starting point for every dispute review.
- Keep a human administrator in control of the final decision.

6. AI-Assisted ID Verification Pre-Screening (Multimodal Vision-Language Model) — Stretch Goal

Uploaded national ID documents are pre-screened using a vision-capable LLM, which checks for basic legibility, apparent tampering, and whether the document type matches what was requested, before the document reaches the admin for final manual approval. This does not replace human verification but reduces the number of clearly invalid submissions administrators need to review manually.

This feature enables the platform to:

- Reduce administrator workload by filtering out clearly invalid or illegible submissions before manual review.
- Speed up the overall worker verification turnaround time.
- Retain a mandatory human-in-the-loop approval step for final verification decisions.

1.6 Project Deliverables

- A fully functional, deployed web-based QuickHire platform with Client, Worker, and Admin roles.
- Worker verification system with national ID upload and admin approval workflow.
- Two-way rating and review system with sentiment analysis output on worker profiles.
- AI-powered Smart Worker Recommendation Engine integrated into the job posting flow.
- Smart Pay Rate Suggestion tool on the job creation form.
- REST API layer to support future mobile app development without backend changes.
- Figma UI/UX prototype designs for all key user flows.
- Full FYP documentation including SRS, sprint reports, architecture diagrams, and traceability matrices.
- Live demonstration and final presentation.

2. Domain Analysis

2.1 Customer

QuickHire does not have a contracted external client organization. The platform is developed as a self-initiated Final Year Project and is designed to serve Pakistan's general public — specifically households and families in urban centres (initially Lahore), event organizers, small businesses, and the informal labour workforce of daily-wage workers and domestic helpers across Pakistan.

The platform addresses a market segment that is currently unserved by any existing digital solution in Pakistan: short-term, flexible, and accountable labour hiring. The target client base ranges from a Lahore household needing a domestic helper for Eid week, to a wedding planner needing event staff for a three-day function, to a family needing a certified nurse post-surgery.

2.2 Stakeholders

The table below lists all ten stakeholders for the QuickHire platform, including both internal actors (inside the system boundary) and external actors (outside the system boundary). Each entry includes a definition of the stakeholder and the key user stories that drive their interaction with the system.

Legend: ■ Blue rows = Inside system boundary ■ Yellow rows = Outside system boundary

Table 1: List of Stakeholders — QuickHire

Stakeholder	Definition & User Stories
1. Client / Hirer [Inside system boundary]	A household, individual, or business that posts jobs and books short-term workers across QuickHire's 14 service categories. User Stories: - As a client, I want to post a job with category, location, date, and pay rate so that suitable workers can apply. - As a client, I want to browse AI-recommended worker profiles so that I can choose the best match. - As a client, I want to confirm, reschedule, or cancel a booking so that I can manage my schedule flexibly. - As a client, I want to rate and review a worker after job completion so that I help future clients decide. - As a client, I want to pay securely through the platform so that I never handle cash with an unknown worker.
2. Individual Worker [Inside system boundary]	A solo service provider — nurse, cook, driver, guard, gardener, childcare or elder care giver — offering skills across the 14 categories. User Stories: - As a worker, I want to create a verified profile with skills, certifications, and availability so that clients can trust and hire me. - As a worker, I want to browse and apply for posted jobs in my category so that I find matching work. - As a worker, I want to receive an AI-suggested pay rate so that I price myself competitively. - As a worker, I want to track my booking history and earnings so that I can manage my income. - As a worker, I want to rate my client after a job so that the platform maintains a trustworthy community.
3. Group / Team Worker [Inside system boundary]	A crew or firm — event staffing groups, security firms, moving and delivery teams — that registers collectively and bids on large-scale assignments.

Stakeholder	Definition & User Stories
	User Stories: - As a group, I want to register a team profile with capacity, past work, and member roster so that clients evaluate us for large jobs. - As a group, I want to bid on event staffing, security deployments, or moving assignments so that we secure group-scale work. - As a group, I want to manage internal member availability and job assignments so that our operations stay coordinated. - As a group, I want to receive both group-level and individual-member ratings so that our reputation is accurately reflected.
4. Site Visitor [Inside system boundary]	A person browsing QuickHire without registering — exploring service categories, reading worker profiles, or evaluating the platform before signing up. User Stories: - As a site visitor, I want to browse available service categories so that I understand what QuickHire offers before registering. - As a site visitor, I want to see sample worker profiles and ratings so that I can assess worker quality. - As a site visitor, I want to see how the hiring process works so that I feel confident enough to create an account.
5. Platform Administrator [Inside system boundary]	A QuickHire operations team member who approves registrations, resolves disputes, configures categories, and generates platform reports. User Stories: - As an admin, I want to approve or reject worker registration requests so that only verified workers appear on the platform. - As an admin, I want to view a live dashboard of bookings, payments, and users so that I can monitor platform health. - As an admin, I want to moderate reviews and resolve disputes so that the platform remains fair and trustworthy. - As an admin, I want to configure service categories, pay rate limits, and geographic zones so that the platform fits local market needs. - As an admin, I want to generate compliance and performance reports so that I can submit them to regulators and stakeholders.
6. System Administrator / IT Support [Inside system boundary]	The technical team managing servers, databases, deployments, access controls, and AI model pipelines for the QuickHire platform. User Stories: - As a system admin, I want to manage user roles and access controls so that only authorised personnel access sensitive functions. - As a system admin, I want to monitor server health and database performance so that I can prevent downtime before it affects users. - As a system admin, I want to deploy updates to the React frontend and Django/Node.js backend so that new features reach users without disruption. - As a system admin, I want to retrain and monitor AI models so that recommendations and pay-rate suggestions remain accurate as data grows.
7. Verification Partner [Outside system boundary]	An external background-check or credential-verification agency that confirms worker identity, criminal record, and professional certifications — critical for nursing, childcare, elder care, and security categories. User Stories:

Stakeholder	Definition & User Stories
	- As a verification partner, I want to receive worker data requests from QuickHire so that I can perform background and credential checks. - As a verification partner, I want to return a verified or rejected status flag so that worker profiles are approved or blocked accordingly.
8. Payment Gateway [Outside system boundary]	An external financial provider (e.g. JazzCash, EasyPaisa, or a bank API) that processes client payments, worker disbursements, refunds, and cancellation charges. User Stories: - As a payment gateway, I want to receive payment requests from QuickHire so that I can process client-to-platform transactions securely. - As a payment gateway, I want to return transaction confirmations or failure codes so that the platform updates booking and payment records in real time.
9. Government / Regulatory Authority [Outside system boundary]	Pakistani labour, health, and safety regulators who receive periodic compliance reports and set the legal framework for short-term labour, nursing licenses, and security service registration. User Stories: - As a regulator, I want to receive periodic compliance reports from the platform so that I can audit labour practices and worker safety standards. - As a regulator, I want to be assured that nursing and security workers hold valid licenses so that public safety is maintained.
10. Search Engine Bot [Outside system boundary]	An automated crawler (Google, Bing) that indexes QuickHire's public pages — service listings, worker profiles, and landing pages — so clients and workers discover the platform through organic search. User Stories: - As a search engine bot, I want to access a sitemap.xml so that all public pages are discoverable and indexed efficiently. - As a search engine bot, I want to read structured metadata on worker and service pages so that relevant search results are ranked correctly.

2.3 Affected Groups with Social or Economic Impact

The following groups are directly impacted by the deployment of QuickHire, linked to the platform's core objectives:

Households and Families (Primary Client Base)

Economic Impact: Families gain a structured, cost-transparent channel for finding short-term domestic workers, eliminating the need for intermediary agents who typically charge informal fees. Access to rated and verified workers reduces the risk of hiring unreliable or unvetted help.

Social Impact: The platform introduces a safety layer for families who hire workers into their homes — particularly for childcare and nursing — through identity verification and public review records.

Informal and Daily-Wage Workers

Economic Impact: Workers gain direct access to a steady stream of job opportunities without dependence on informal networks, agents, or word of mouth. A verified profile and public rating history enables workers to command fairer pay and attract more clients over time.

Social Impact: Workers receive a digital professional identity — a meaningful recognition of their skills and experience in a market that has historically treated them as anonymous, replaceable labour.

Event Organizers and Small Businesses

Economic Impact: Organizations that require short-term or event-based staffing gain a reliable, accountable, and searchable workforce directory. This reduces the logistical burden and cost of staffing events and short-term projects.

Social Impact: Formalizing event staffing through a rated and verified platform improves the reliability of services at weddings, corporate events, and similar functions.

Pakistan's Informal Labour Economy (Macro Impact)

Economic Impact: QuickHire contributes to formalizing a segment of the economy that is entirely cash-based and undocumented. Workers building a verifiable digital work history may have an easier path to financial inclusion over time.

Social Impact: By creating a transparent, accountable marketplace, QuickHire challenges the opacity and power imbalance that currently characterizes Pakistan's informal domestic labour sector.

2.4 Dependencies / External Systems

QuickHire is a self-contained web platform, but the following external systems and services are either depended upon or interfaced with:

Email Service Provider (SMTP)

Purpose: Automated email notifications for booking confirmations, reminders, and status updates. **Interaction:** Django's email backend connects to an SMTP provider such as SendGrid or Gmail SMTP for outbound email delivery.

Cloud Hosting Platform

Purpose: Hosting the web application and database for live demonstration. **Interaction:** The Django application and PostgreSQL database are deployed on Heroku or Railway.app. File uploads (worker IDs, documents) are stored using Django's media file handling.

National ID Verification (Manual)

Purpose: Workers upload their national identity documents for identity verification. **Interaction:** The platform does not integrate with NADRA or any external verification API. Admin staff manually review and approve uploaded documents within the platform's admin dashboard.

LLM Provider API and Vector Database (AI/ML)

Purpose: Powering the LLM-Powered Job Description Parser, AI-Powered Natural-Language Worker Search, LLM-Based Review Sentiment and Theme Analysis, AI Support and Onboarding Assistant, and AI Dispute-Summary Assistant. **Interaction:** The Django backend calls an LLM provider (e.g., Groq, OpenAI, or Anthropic) over HTTPS through a dedicated AI service layer, and stores/queries worker-profile embeddings in a vector database (e.g., pgvector, Chroma, or Pinecone) for semantic search.

Git / GitHub

Purpose: Version control and code baseline management. **Interaction:** The development team maintains a branching strategy: main, development, and feature branches, with all code changes managed through pull requests.

2.5 Related Projects with Feature Comparison

2.5.1 Related Projects

The following platforms were studied as reference and comparative material during the research and design phase of QuickHire:

Urban Company (India)

Pakistan's closest international analogue. Urban Company connects households with home service professionals for cleaning, beauty, plumbing, and similar tasks. It is primarily a browse-and-book model for single-session services and does not address the multi-day or event hiring use case. Its worker-side onboarding and rating system directly inspired QuickHire's approach to worker profiles.

GigSmart (USA)

A US-based platform for short-term gig staffing, primarily serving the business sector. Workers browse available shift postings and apply. The model is directly comparable to QuickHire's post-and-apply flow, though GigSmart does not serve household or domestic hiring and is not designed for South Asian labour market conditions.

Rozee.pk (Pakistan)

Pakistan's largest job portal. Designed exclusively for permanent and long-term employment, Rozee.pk has no infrastructure for short-term, per-day, or event-based hiring. Workers can apply to listings but have no booking system, availability calendar, or two-way rating mechanism.

Mahir Company (Pakistan)

The most commonly cited domestic competitor. Mahir connects clients with home service providers for maintenance and beauty appointments. It uses a browse-and-book model for single-session services only and does not support multi-day, event, or domestic care hiring. Worker identities are not individually visible — the platform brand is primary.

Instawork (USA / Canada)

A platform for temporary business staffing — restaurants, warehouses, and hospitality venues. The shift-based model is relevant to QuickHire's event staffing use case, but Instawork does not serve household hiring and is designed for developed market labour conditions.

2.5.2 Feature Comparison with Pakistan Competitors

Table 2: Feature Comparison — QuickHire vs. Pakistan Platforms

Platform	Short-Term?	Event Hiring?	Worker Applies?	Multi-Category?	Worker Profile
Mahir Company	No	No	No	Partial	No
Karsaaz	No	No	No	Partial	No
Rozee.pk	No	No	Yes	Yes	Basic
OLX Pakistan	No	No	No	Yes	No
Facebook Groups	No	No	No	Yes	No
QuickHire	Yes	Yes	Yes	Yes	Full Verified

2.5.3 How QuickHire Differs from Mahir Company

Table 3: Detailed Feature Comparison — QuickHire vs. Mahir Company

Feature	Mahir Company	QuickHire
Booking Model	Browse services, book a slot	Post a job, workers apply
Duration Focus	Single task (1–2 hours)	Hours, days, weeks, events
Service Categories	Home maintenance & beauty	14 categories incl. childcare, nursing, events
Worker Identity	Platform brand, anonymous worker	Full individual profile & digital career
Rating System	Client rates worker only	Two-way: both sides rate each other
Target Market	Home repair & salon services	Domestic, care, events, construction
Worker Verification	Not required	National ID upload & admin approval

2.5.4 Global Landscape — Where QuickHire Fits

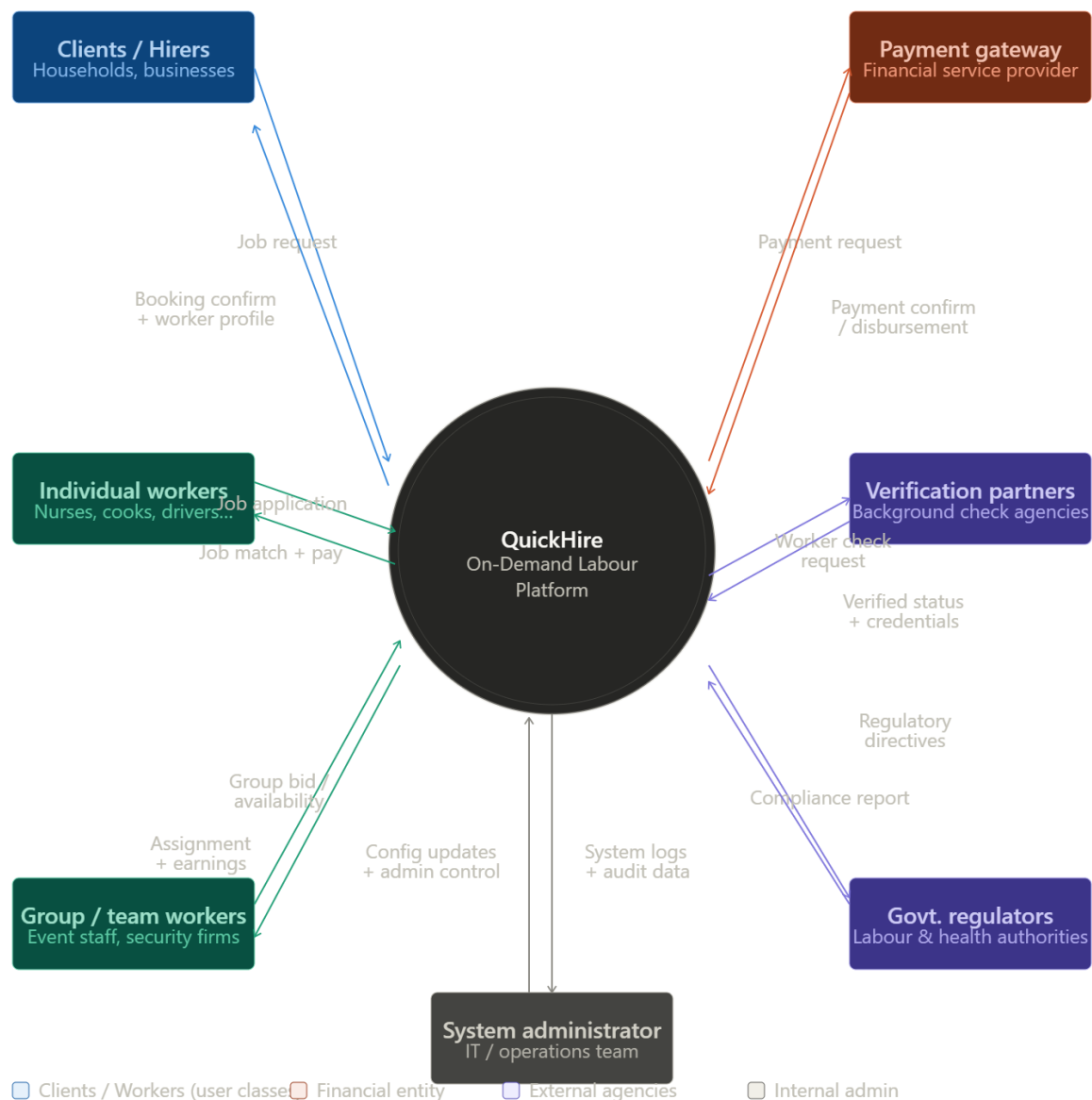
Table 4: Global Platform Comparison

Platform	Country	Model	Household Hiring?	Short-Term Focus?
Urban Company	India	Browse & book	Yes	Partial
GigSmart	USA	Post & apply	No	Yes
Instawork	USA/Canada	Business shifts	No	Yes
Coople	Europe	Temp staffing	No	Yes
Jobble	USA	Browse shifts	No	Yes
QuickHire	Pakistan	Post & apply	Yes	Yes

QuickHire is the only platform combining household-level domestic hiring with a short-term job-posting model specifically designed for Pakistan's informal South Asian labour market. No existing platform — globally or domestically — addresses the combination of short-duration bookings, household domestic hiring, a post-and-apply worker model, and a two-way verified rating system within a single Pakistani-market product.

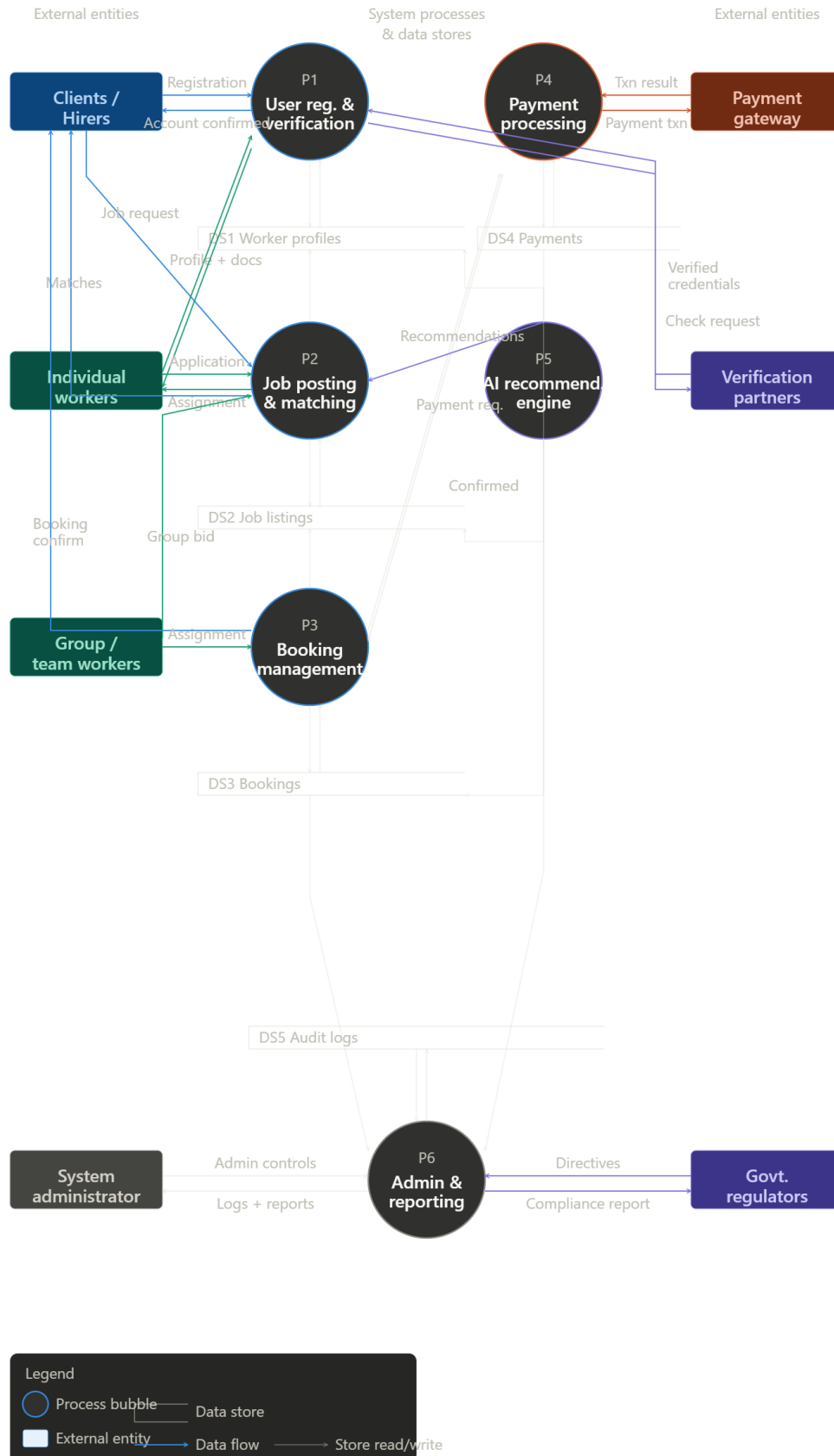
2.6 Context Diagram

The context diagram for QuickHire illustrates the system boundary and identifies all external entities that interact with the platform. The entire QuickHire system is represented as a single circle, with no internal visibility. External entities include Clients (households, event organizers, businesses), Workers (domestic helpers, skilled labourers, caregivers), the Platform Administrator, the Email Service Provider (for automated notifications), and the Cloud Hosting Platform (Heroku/Railway.app). Data flows include: job postings and booking requests from Clients to the system; job applications and profile data from Workers to the system; worker verification approvals from the Admin to the system; and automated email confirmations from the system to both Clients and Workers.



2.7 Data Flow Diagram Level 0

The Level 0 DFD expands the QuickHire system context into its six major processes: (1) User Registration and Verification, where workers upload national IDs and await admin approval; (2) Job Posting, where clients submit job listings with category, city, duration, and pay rate; (3) Worker Matching and Recommendation, where the AI engine scores and ranks workers for each job post; (4) Application and Booking Management, where workers apply and bookings progress through the Pending → Confirmed → Completed lifecycle; (5) Rating and Review, where both parties submit post-job ratings and the sentiment analysis module processes written reviews; and (6) Admin Operations, where the administrator manages verifications, disputes, and platform analytics.



3. Requirements Analysis

3.1 List of Actors

The following actors interact with the QuickHire platform. All actors are also represented in the Stakeholders table in Section 2.2.

Client: A registered individual, household, event organizer, or business that posts job listings and books workers. The Client interacts with job creation, worker profile browsing, booking management, and the post-job rating system.

Worker: A registered individual from Pakistan's informal labour pool who has uploaded their national ID and been verified by the admin. The Worker browses and applies to posted jobs, manages their availability calendar, and builds a public profile with ratings and job history.

Platform Administrator: A privileged system user who manages all accounts, reviews and approves worker verifications, monitors booking activity, resolves disputes between clients and workers, and accesses platform-wide analytics.

Email Service (External Actor): The SMTP-based email system that receives automated notification requests from QuickHire and delivers booking confirmations, reminders, and status update emails to registered Clients and Workers.

AI/ML Engine (Internal System Actor): The LLM-based AI service layer that parses job descriptions, performs retrieval-augmented worker matching, analyses review sentiment and themes, powers the support assistant, and generates dispute summaries for the admin.

3.2 Product Backlog

The Product Backlog for QuickHire is organized into Epics representing the major functional areas of the platform. Each Epic is broken down into User Stories with full acceptance criteria, story points, priority, and status. All backlog items follow the format: PB-ID | User Story | Priority | Acceptance Criteria | Story Points | Status.

3.2.1 User Registration and Verification

Table 5: Product Backlog — Epic: User Registration and Verification

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
RV-01	As a new client, I can register an account with my name, email, and city so that I can post jobs and book workers.	High	Client registration form accepts valid inputs; confirmation email is sent; account is accessible immediately.	3	Pending
RV-02	As a new worker, I can register an account with my name, email, city, and service category so that I can apply to jobs.	High	Worker account is created; service category is saved; worker is directed to profile setup page.	3	Pending
RV-03	As a worker, I can upload my national identity document so that I can be verified by the admin and become eligible to apply for jobs.	High	Worker can upload CNIC image or PDF; upload is stored securely; status shows 'Pending Verification'.	5	Pending
RV-04	As an admin, I can review pending worker verification requests and approve or	High	Admin dashboard lists all pending verifications; approve/reject actions	3	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
	reject them so that only verified workers appear on the platform.		update worker status; worker receives email notification.		
RV-05	As a registered user, I can log in and log out securely with role-based redirection so that I access only my permitted features.	High	Login enforces correct role redirect (Client → client dashboard, Worker → worker dashboard, Admin → admin panel); unauthorized access returns 403.	2	Pending
RV-06	As a user, I can reset my password via a secure email link so that I can recover my account if I forget my credentials.	Medium	Password reset email is sent within 2 minutes; link expires after 24 hours; new password is saved and login works immediately.	3	Pending
RV-07	As an admin, I can manually suspend or reactivate any user account so that I can enforce platform safety policies.	Medium	Suspended users cannot log in; account shows 'Suspended' status in admin panel; reactivation restores access immediately.	2	Pending
RV-08	As a returning user, I can update my email address or phone number so that my contact details remain current.	Low	Updated contact info is saved; a confirmation email is sent to the new address before the change takes effect.	2	Pending

3.2.2 Worker Profile Management

Table 6: Product Backlog — Epic: Worker Profile Management

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
WP-01	As a worker, I can build a detailed profile including a photo, bio, years of experience, and service category so that clients can evaluate me.	High	Profile fields are saved and publicly visible; photo uploads are accepted in JPG/PNG formats up to 2 MB.	3	Pending
WP-02	As a worker, I can set and update my availability calendar so that I am only shown as bookable on dates I am free.	High	Calendar reflects unavailable dates; system prevents clients from booking on blocked dates; worker can update at any time.	5	Pending
WP-03	As a worker, I can view my complete job history, including past bookings and ratings, so that I can track my platform reputation.	Medium	Job history lists all past and current bookings with status, dates, and client ratings.	3	Pending
WP-04	As a worker, I can specify my daily or hourly rate expectation so that clients know my pricing before booking.	Medium	Rate is visible on worker profile card; clients can filter workers by rate range.	2	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
WP-05	As a worker, I can list multiple service sub-categories (e.g., both Cooking and Cleaning) so that I appear in more relevant job searches.	Medium	Multiple categories can be selected during profile setup; worker appears in search results for all selected categories.	3	Pending
WP-06	As a client, I can view a worker's full verified profile including experience, rating, reviews, and availability before booking so that I can make an informed decision.	High	Worker profile page loads all sections; verified badge is shown; availability calendar is visible; reviews are listed newest first.	3	Pending
WP-07	As a worker, I can receive a verified badge on my profile upon admin approval of my national ID so that clients can trust my identity.	High	Verified badge appears on profile and search cards after admin approves CNIC; badge is removed if account is suspended.	2	Pending

3.2.3 Job Posting and Browsing

Table 7: Product Backlog — Epic: Job Posting and Browsing

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
JP-01	As a client, I can post a job listing specifying category, city, start date, duration, and pay rate so that interested workers can apply.	High	Job post form requires all mandatory fields; posting saves to DB and appears in worker browse view within 1 minute.	5	Pending
JP-02	As a worker, I can browse all active job listings and filter by category and city so that I can find relevant opportunities.	High	Filter controls update job list without page reload; only active (non-expired, non-filled) listings are shown.	3	Pending
JP-03	As a worker, I can apply to a job listing with a short cover message so that the client is notified and can review my application.	High	Application is saved with cover message; client receives email notification; application appears in client's dashboard.	3	Pending
JP-04	As a client, after posting a job, I can immediately view the top five AI-recommended workers with their suitability scores so that I can proactively reach out to suitable workers.	High	Recommendation engine returns ranked list within 2 seconds via AJAX; score and one-line explanation are displayed per worker.	8	Pending
JP-05	As a client, I can close or delete a job listing once it is filled so that workers do not apply to positions that are no longer available.	Medium	Closed listings do not appear in worker browse view; existing applicants are notified of closure via email.	2	Pending
JP-06	As a client, I can edit a job listing (pay rate, duration, or description) before any	Medium	Edit form is accessible from the client dashboard; changes are saved	2	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
	worker has been confirmed so that I can correct mistakes.		and the updated listing is immediately visible.		
JP-07	As a worker, I can save a job listing to my shortlist so that I can apply to it later without losing track of it.	Low	Shortlisted jobs appear in a dedicated 'Saved Jobs' section in the worker dashboard; shortlist persists across sessions.	2	Pending
JP-08	As a client, I can view all applications received for a job listing in one place so that I can compare candidates efficiently.	High	Applications page lists all applicants with their profile summaries, cover messages, and application timestamps.	3	Pending

3.2.4 Booking Management

Table 8: Product Backlog — Epic: Booking Management

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
BK-01	As a client, I can confirm a worker's application to change the booking status from Pending to Confirmed so that both parties are notified by email.	High	Booking status updates to Confirmed; confirmation email is sent to both client and worker within 1 minute.	3	Pending
BK-02	As a client, I can mark a completed booking so that the status changes to Completed and the rating flow is triggered.	High	Booking status updates to Completed; both parties receive a rating prompt via email and in-app notification.	3	Pending
BK-03	As a worker, I can view all my current and past bookings with their status so that I can manage my schedule.	High	Booking list shows status labels (Pending, Confirmed, Completed, Cancelled); sorted by start date.	3	Pending
BK-04	As a client, I can cancel a confirmed booking before the start date so that the worker is notified and the slot is freed.	Medium	Cancellation updates booking status to Cancelled; worker receives email notification; job listing can be reopened.	3	Pending
BK-05	As a worker, I can decline a booking request so that I am not obligated to accept every confirmed application.	Medium	Declined bookings return to Pending status for the client; client is notified and can confirm a different applicant.	2	Pending
BK-06	As a client, I can re-book a worker I have previously hired directly from my booking history so that I can skip the job posting step for repeat hires.	Low	Re-booking form is pre-filled with previous booking details; worker receives a direct booking request notification.	3	Pending
BK-07	As a worker, I can set a status to 'Currently Engaged' so that no new bookings are	Medium	Worker's profile shows 'Currently Engaged' badge; workers with this	2	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
	made during an active assignment.		status are excluded from AI recommendations.		

3.2.5 Rating and Review System

Table 9: Product Backlog — Epic: Rating and Review System

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
RR-01	As a client, after a booking is marked Completed, I can submit a star rating and written review for the worker so that future clients are informed.	High	Rating (1–5 stars) and review text are submitted; saved to DB; visible on worker profile within 1 minute.	3	Pending
RR-02	As a worker, after a booking is marked Completed, I can submit a star rating and written review for the client so that future workers are informed.	High	Worker rating of client is saved; visible to other workers who view that client's profile.	3	Pending
RR-03	As any user viewing a worker profile, I can see the worker's average star rating, review count, and extracted sentiment keywords from their reviews.	High	Average rating and review count are displayed on profile header; sentiment keywords are listed in a 'What clients say' section.	5	Pending
RR-04	As a client, I can flag a review as inappropriate so that the admin can moderate harmful content.	Medium	Flag button appears on each review; flagged reviews are queued in the admin moderation panel.	2	Pending
RR-05	As a worker, I can respond to a client review on my profile so that I can address feedback publicly.	Low	Worker can post one reply per review; reply is shown below the original review.	2	Pending
RR-06	As an admin, I can remove a review that violates platform policies so that the platform maintains accurate and fair content.	Medium	Admin can delete any review from the moderation panel; deletion is logged with reason; both parties are notified.	2	Pending

3.2.6 Admin Dashboard

Table 10: Product Backlog — Epic: Admin Dashboard

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
AD-01	As an admin, I can view platform-wide analytics including total users, active jobs, completed bookings,	High	Analytics dashboard loads within 3 seconds; metrics are accurate and updated in real-time or with a maximum 5-minute lag.	5	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
	and pending verifications so that I can monitor platform health.				
AD-02	As an admin, I can view, suspend, or deactivate any Client or Worker account so that I can maintain platform quality and resolve disputes.	High	Admin account management panel lists all users; suspend/deactivate actions take effect immediately.	3	Pending
AD-03	As an admin, I can manage all job listings and bookings and intervene in disputes so that the platform remains fair and accountable.	High	Admin can view, edit, or delete any job listing or booking; dispute resolution notes can be logged against a booking.	3	Pending
AD-04	As an admin, I can send a platform-wide announcement to all registered users via email so that I can communicate policy updates or maintenance notices.	Medium	Announcement email is dispatched to all active users within 10 minutes; admin can preview before sending.	3	Pending
AD-05	As an admin, I can view and export platform usage reports (registrations, bookings, completions) as CSV so that I can share data for academic or evaluation purposes.	Medium	CSV export includes all required fields; file is generated within 5 seconds; data is accurate as of export time.	3	Pending
AD-06	As an admin, I can moderate flagged reviews and take action (remove or clear flag) so that review content remains fair and accurate.	Medium	Flagged review queue is accessible from admin panel; admin can approve or delete with a reason log.	2	Pending

3.2.7 AI and Smart Features

Table 11: Product Backlog — Epic: AI and Smart Features

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
AI-01	As a client, when I view a job I have posted, I can see a ranked list of the top five most suitable available workers with their suitability score and a one-line explanation so that I can make an informed hiring decision.	High	Recommendation list loads via AJAX in under 2 seconds; suitability score and explanation are shown per worker; list updates when filters change.	8	Pending
AI-02	As a client, when I create a job posting, I can see a suggested fair pay rate range based on category, city, and duration so that I can set a competitive and reasonable rate.	High	Pay rate suggestion appears in the job form within 1 second of selecting category and city; range is formatted as 'Rs. X–Y per day'.	5	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
AI-03	As any user viewing a worker profile, I can see automatically extracted sentiment keywords such as 'punctual', 'reliable', or 'experienced' from the worker's reviews so that I can quickly assess the worker's strengths.	High	Sentiment keywords are updated after each new review; top 5 positive keywords are displayed in a visual tag list on the profile.	5	Pending
AI-04	As a worker, I can see a suggested profile completion score so that I know what to add to attract more bookings.	Low	Profile score (0–100%) appears on the worker dashboard; tips for improvement are shown for incomplete fields.	3	Pending
AI-05	As an admin, I can view a dashboard widget showing AI model performance metrics so that I can monitor recommendation accuracy.	Low	Dashboard shows recommendation click-through rate and booking conversion rate from AI-recommended workers.	3	Pending

3.2.8 Non-Functional Requirements

Table 12: Product Backlog — Non-Functional Requirements

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
NFR-01	The platform shall be fully mobile-responsive and functional on screens as small as 375px width without loss of core functionality.	High	All pages render correctly on 375px, 768px, and 1280px viewports; no horizontal scroll on mobile; buttons and forms are usable on touch screens.	5	Pending
NFR-02	The system shall enforce role-based access control, ensuring that Client, Worker, and Admin roles can only access their respective features.	High	Unauthorized role access returns HTTP 403; all protected routes are tested for each role combination.	3	Pending
NFR-03	The REST API layer shall be implemented using Django REST Framework so that all core features can be accessed programmatically for future mobile app development.	High	API endpoints return JSON with standard HTTP status codes; authentication uses JWT tokens; Swagger/DRF browsable API is accessible.	5	Pending
NFR-04	All authentication shall use secure session management with JWT support for API access.	High	JWT tokens expire in 24 hours; refresh tokens are issued; all API routes require valid token; no sensitive data is stored in localStorage.	3	Pending
NFR-05	The platform shall deliver AI-powered worker recommendations in under two seconds via asynchronous AJAX calls.	High	Recommendation API endpoint responds within 2 seconds for up to 1,000 worker profiles; load tested and verified.	3	Pending

PB-ID	User Story	Priority	Acceptance Criteria	Story Points	Status
NFR-06	The system shall maintain 99% uptime during the FYP evaluation and demonstration period.	Medium	Deployment on Heroku/Railway with health check monitoring; downtime incidents are logged and resolved within 1 hour.	2	Pending
NFR-07	All user-uploaded files (CNIC images, profile photos) shall be stored securely and not publicly accessible via direct URL.	High	Files are stored with randomized filenames; direct URL access returns 403; only authenticated admin can view CNIC uploads.	3	Pending

3.3 Figma UI/UX Designs

This section contains the Figma prototype designs for all key user flows in the QuickHire platform. The designs cover the complete journey for each of the three roles: Client, Worker, and Admin. All wireframes and high-fidelity mockups are developed in Figma and are accessible via the link below.

The following screens are included in the Figma prototype:

- PID-01: User Registration Page (Client / Worker)

Join QuickHire Today

Connect with thousands of verified workers across Pakistan

- Verified profiles and secure payments
- AI-powered matching for best results
- 24/7 customer support

Create Account

Get started with QuickHire

Client

Worker

Full Name

Email

Phone

City

Account Type

Individual/Family

Password

Confirm Password

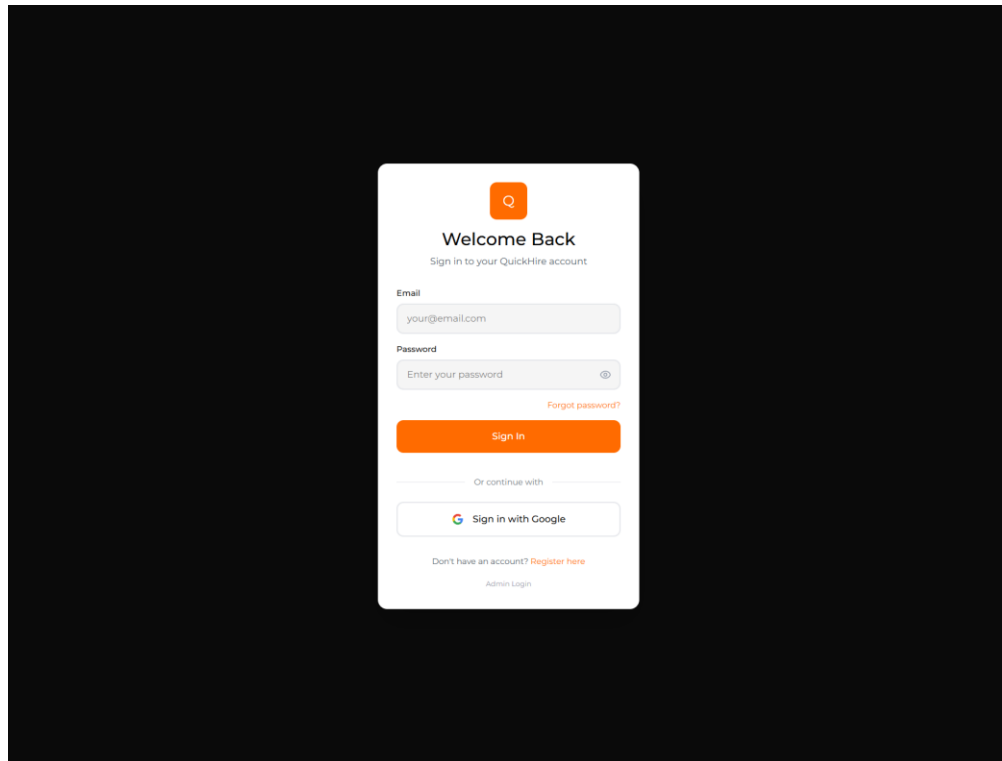
Create Account

Or continue with

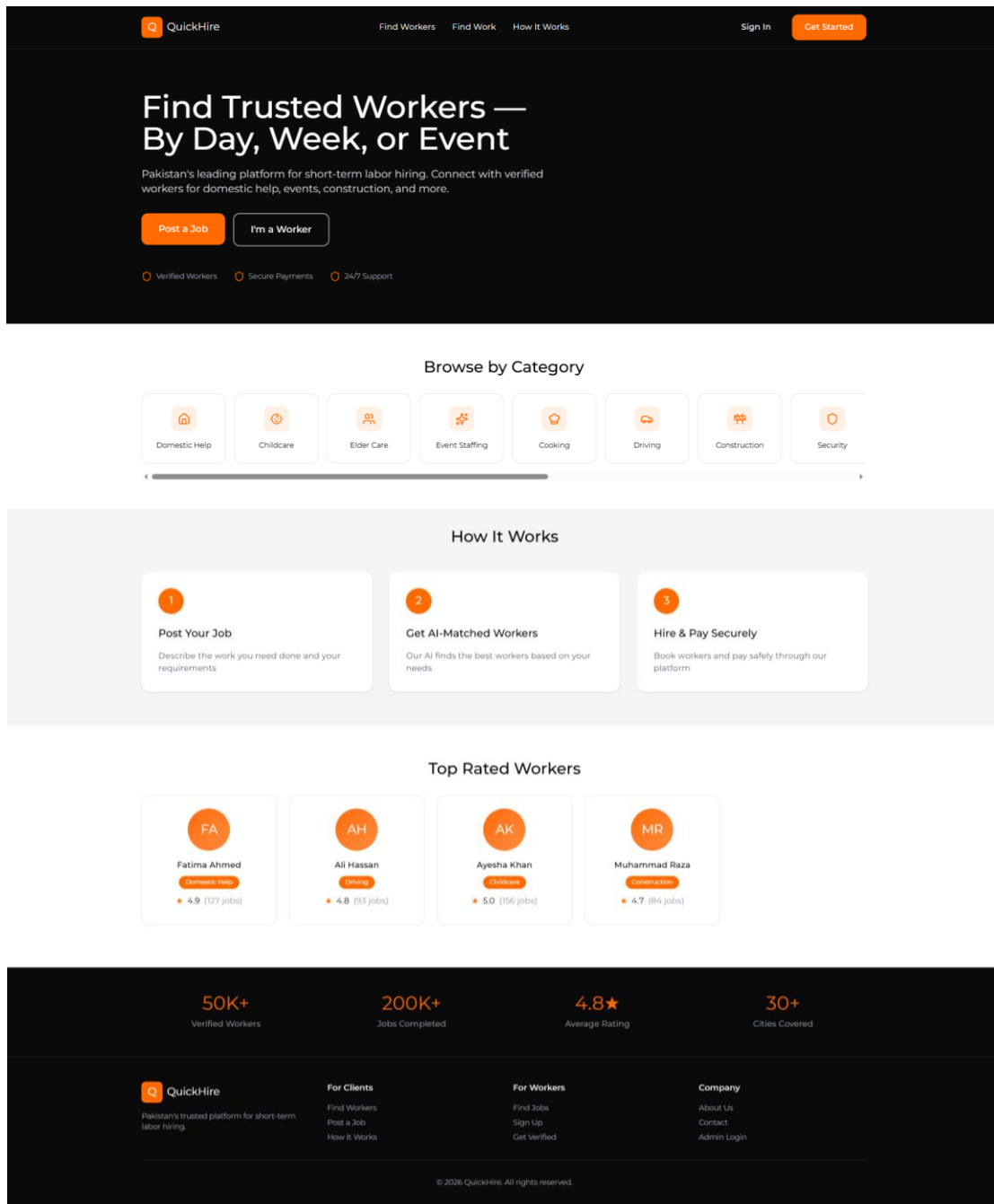
Sign up with Google

Already have an account? [Sign in here](#)

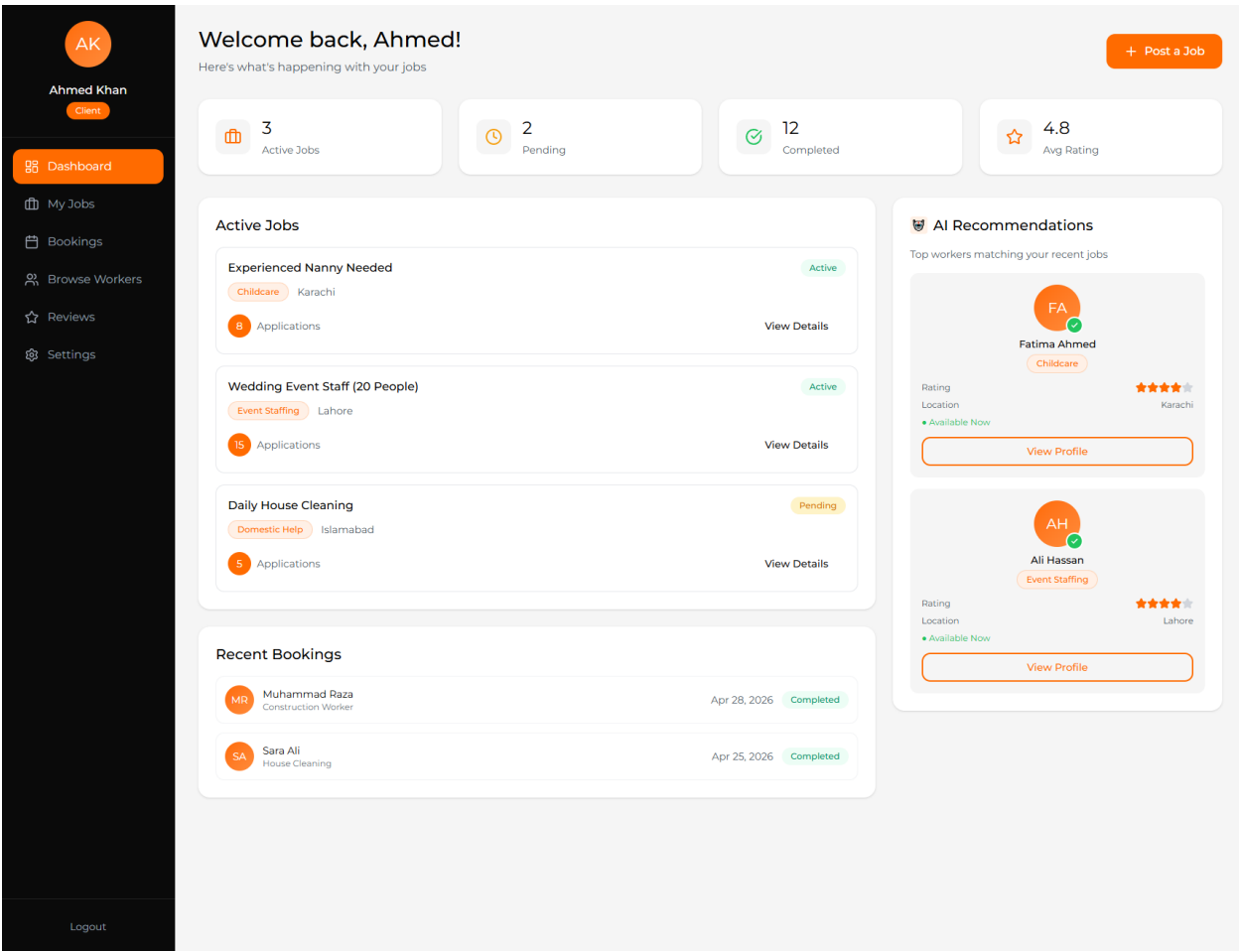
- PID-02: User Login Page (Client / Worker Authentication)



- PID-03: Home / Landing Page



- PID-04: Client Dashboard — Overview & Activity Panel



- PID-05: Job Posting Page — Create New Job Listing

AK

Ahmed Khan

Client

Dashboard

My Jobs

Bookings

Browse Workers

Reviews

Settings

Logout

Post a New Job

Fill in the details to find the perfect workers

Job Details

Job Title

e.g., Experienced Nanny Needed

Category

Select a category

Job Description

Describe the work required, expectations, and any special requirements...

Location & Schedule

City

Select city

Area/Neighborhood

e.g., DHA, Gulberg

Start Date

dd/mm/yyyy

End Date

dd/mm/yyyy

Duration Type

Days

Weeks

Event

Number of Workers

-

1

+

Pay Rate

Pay Rate (PKR per day)

e.g., 1500-2000

AI Suggested Rate

PKR 1,500 – 2,000/day

Based on the category, location, and current market rates

Use Suggested Rate

Requirements

Required Skills (comma separated)

e.g., CPR Certified, Early Education, Meal Prep

Gender Preference

Any

Additional Notes

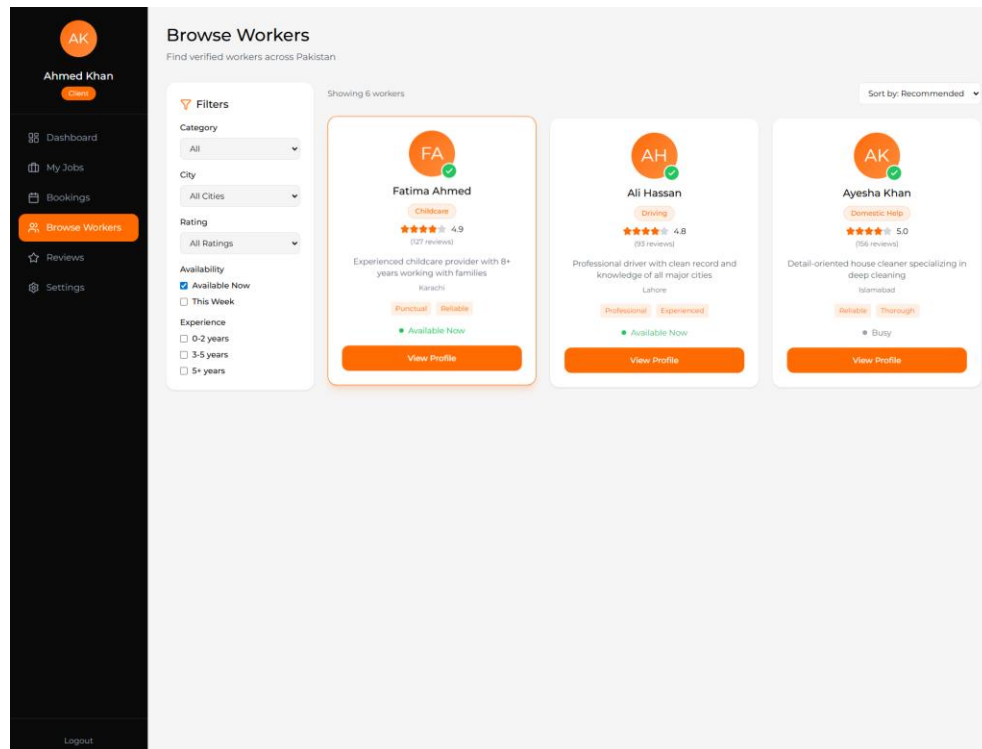
Any other requirements or information...

Preview

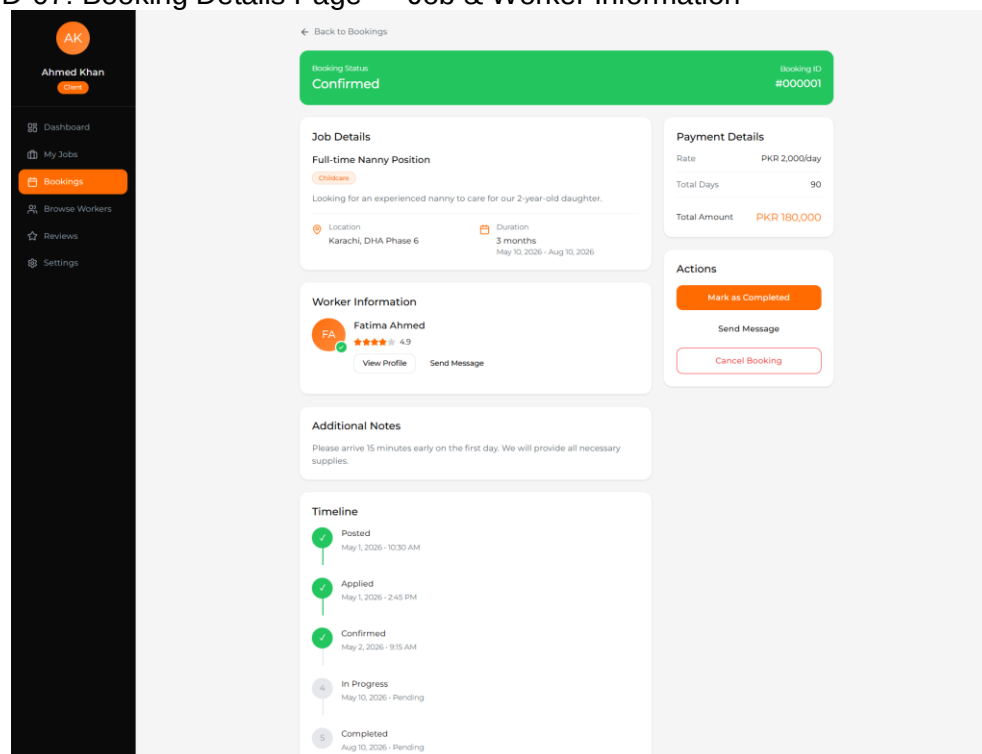
Post Job Now

Page 26 of 43

- PID-06: Browse Workers Page — Search & Filter



- PID-07: Booking Details Page — Job & Worker Information



- PID-08: Rating & Review Submission Page

← Back

Leave a Review

Fatima Ahmed
Childcare
Full-time Nanny Position

How would you rate your experience?

★★★★★
Select a rating

Tell us more about your experience

Share details about your experience with this worker...

0/500 characters

Quick tags (select all that apply)

Punctual Reliable Professional Friendly Skilled Hardworking
Late Unprofessional Not Recommended

Cancel Submit Review

Your review helps other clients make informed decisions and helps workers improve their services.
Reviews are public and cannot be edited once submitted.

- PID-09: Worker Dashboard — Jobs & Booking Management

Fatima Ahmed
Worker

Dashboard

Find Jobs
My Jobs
Earnings
Reviews
Settings

Logout

Your account is verified!
You can now apply for jobs and receive bookings.

Welcome back, Fatima!

Here's your work overview

5 Applications
2 Active Jobs
24 Completed
4.9 Rating

Recommended Jobs

View All

Full-time Nanny Position
Childcare Family Karachi
PKR 2,000/day · 3 months · 2 hours ago
Apply Now

Weekend Childcare
Childcare Individual Karachi
PKR 1,800/day · Ongoing · 5 hours ago
Apply Now

Event Childcare - Wedding
Event Staffing Event Organizer Lahore
PKR 2,500/day · 1 day · 1 day ago
Apply Now

Upcoming Bookings

Daily Childcare Confirmed
Ahmed Family
May 5, 2026 · 9:00 AM - 5:00 PM

Weekend Care Pending
Khan Family
May 10, 2026 · 10:00 AM - 6:00 PM

Boost Your Profile

Complete your profile to get more job matches

- Add work samples
- Update availability
- Add more skills

Complete Profile

Quick Stats

Profile Views: 234
Response Rate: 98%
Job Success: 96%
Total Earnings: PKR 48,000

Recent Activity

New job match: 2 hours ago
Booking confirmed: 5 hours ago
New review received: 1 day ago

- PID-10: Worker Profile Page — Skills, Ratings & Details

FA

FA

Childcare

Verified

4.9

(127 reviews)

8 years experience

Message

Book Now

About

Experienced and caring childcare provider with over 8 years of experience working with families across Karachi. Specialized in newborn care, toddler development, and educational activities. CPR certified and background checked. I believe in creating a safe, nurturing environment where children can learn and grow.

Quick Info

Location

Karachi

Experience

8 years

Success Rate

98%

AI Sentiment Analysis

Review Analysis

Based on 127 reviews, here are the most common themes:

Punctual (92%)

Reliable (88%)

Professional (85%)

Caring (90%)

Overall Sentiment: Very Positive

Recent Jobs

Full-time Nanny

★★★★★ 5.0

Khan Family

Jan 2024 - Present

Weekend Childcare

★★★★☆ 4.8

Ahmed Family

Nov 2023 - Dec 2023

Event Childcare

★★★★★ 5.0

Wedding Event

Oct 2023

Reviews (127)

SK

Sarah Khan

★★★★★ Apr 15, 2026

Fatima is absolutely wonderful with our two kids. Very punctual and professional. Highly recommend!

Punctual

Reliable

Professional

AA

Ali Ahmed

★★★★★ Mar 22, 2026

Great experience overall. Our daughter really enjoyed spending time with Fatima.

Friendly

Experienced

ZM

Zainab Malik

★★★★★ Feb 10, 2026

Best childcare provider we have ever hired. Will definitely book again!

Reliable

Caring

Trustworthy

Availability

1

2

3

4

5

6

7

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10

11

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16

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Today

Available

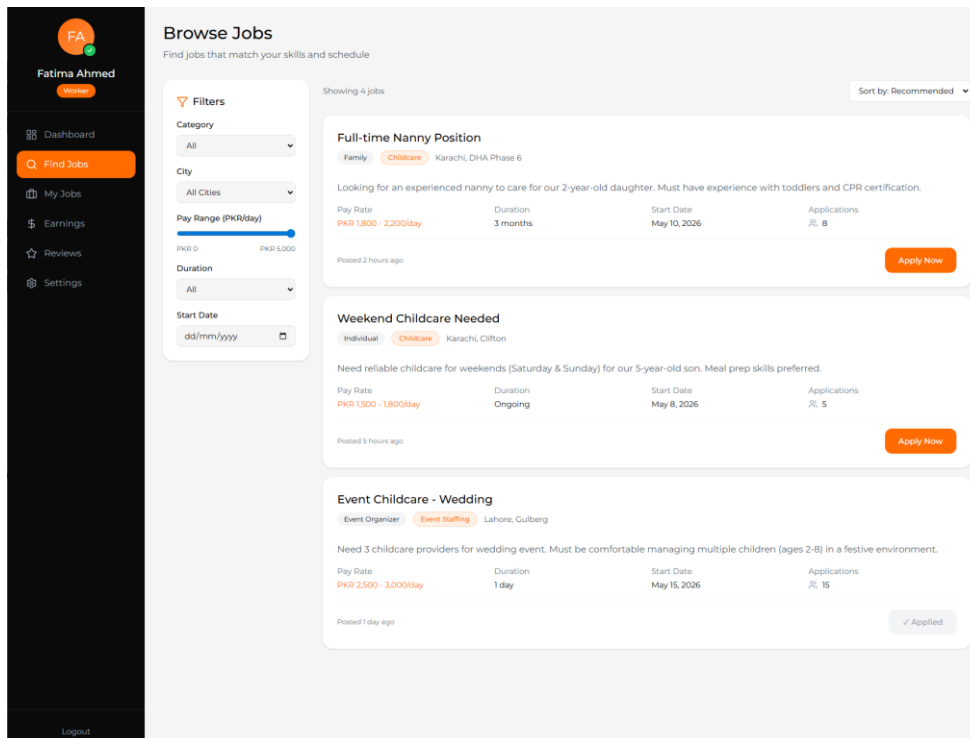
Busy

Ready to hire?

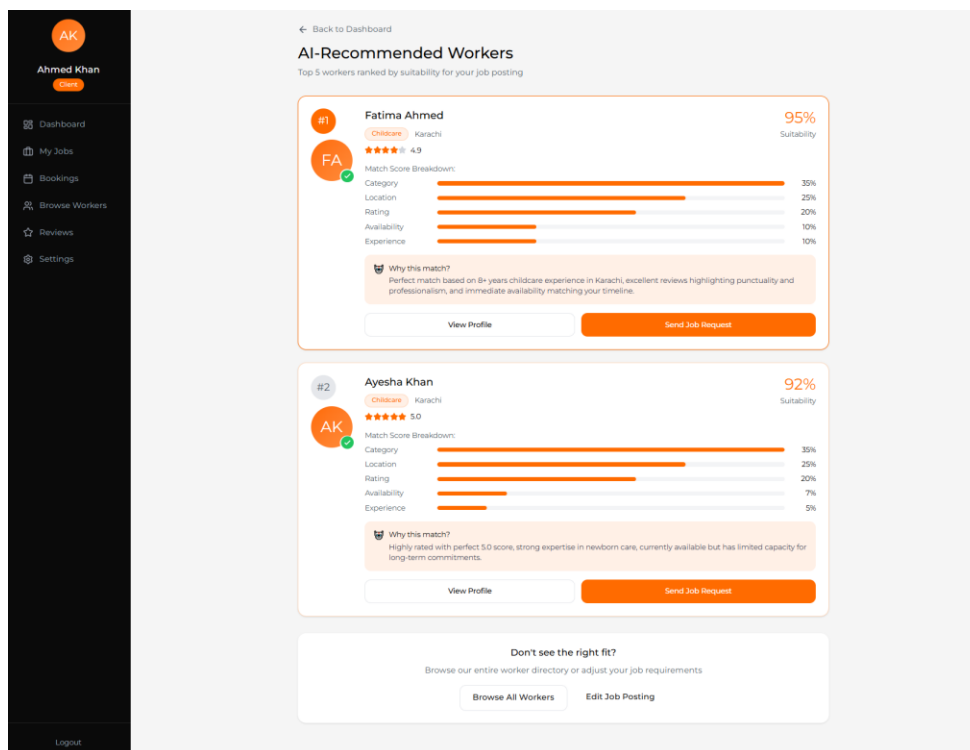
Book Fatima for your next job

Book Now

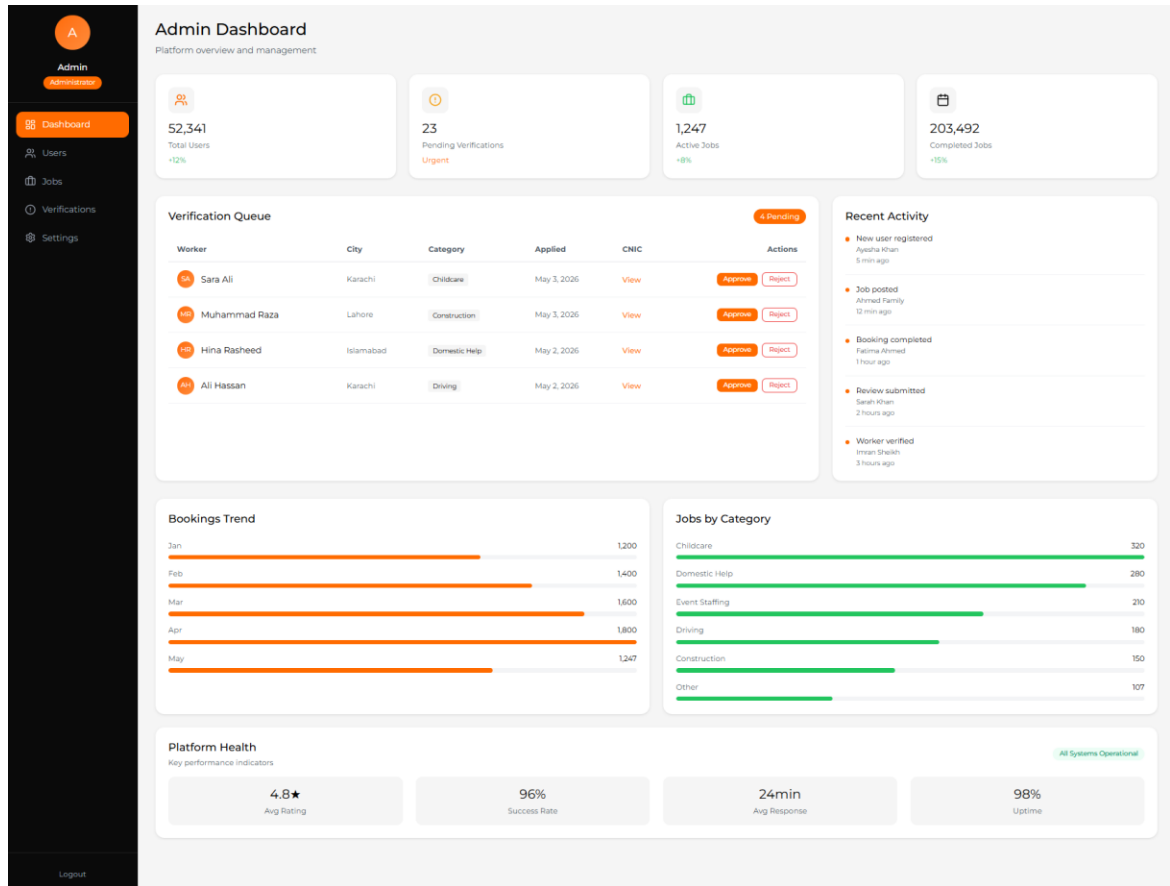
- PID-11: Browse Jobs Page — Available Job Listings



- PID-12: AI Recommendations Page — Suggested Workers / Jobs



- PID-13: Admin Dashboard — System Overview & Controls



4. Project Planning and Execution using Sprints

4.1 Jira

Jira is used for Product Backlog grooming, Sprint 1 backlog execution, Sprint 2 planning, task assignment, story point estimation and status tracking.

4.2 Slack

Slack is used for formal team communication, daily updates, blockers, meeting coordination and sprint discussion evidence instead of WhatsApp.

4.3 GitHub Repository

GitHub is used for version control with main, development and feature branches. Pull requests are used before merging completed features. GitHub Copilot is used for AI-assisted coding, boilerplate generation, test suggestions and code review support.

4.4 Sprint 1

4.4.1 Sprint 1 Planning Meeting Minutes

Field		Details		
Meeting Title		Sprint 1 Planning Meeting - Foundation and Core Marketplace Flow		
Date		Wednesday, 13 May 2026		
Time / Duration		10:00 AM - 11:15 AM / 75 minutes		
Mode		Supervisor meeting and team planning session		
Facilitator		Dr. Ehtesham ul haq Dar / Scrum Master: Ali Husnain		
Attendees		Ali Husnain, Husnain Azam, Muhammad Sheraz		
Sprint Duration		13 May 2026 - 02 June 2026		
Sprint Goal		Establish the QuickHire foundation by completing authentication, worker profile and ID verification, client job posting, worker application flow, and the first working version of the smart worker recommendation engine.		
Definition of Done		Code committed to GitHub, UI reviewed against Figma, functionality tested through documented test cases, Jira task status updated, and no critical defects pending.		
Selected PB-ID		Reason for Selection		Priority
AM-01		Authentication is required before any role-based feature can be used.		Must
AM-02		Basic account/profile data is required for role dashboards.		Should
WP-01		Worker profiles are essential for the marketplace supply side.		Must
WP-02		National ID upload supports the QuickHire trust layer.		Must
AD-01		Admin approval is required before workers can be publicly trusted.		Must
JP-01		Job posting is the primary client-side marketplace action.		Must
BK-01		Worker application enables demand-supply connection.		Must
MT-01		Smart ranking differentiates QuickHire from informal hiring channels.		Must
Member	Days Available	Hours/Day	Total Capacity	Main Responsibility
Ali Husnain	15	5	75	Scrum coordination, Jira/GitHub setup, client job posting, applications, documentation.
Husnain Azam	15	5	75	Backend APIs, authentication, worker verification, recommendation engine, database models.
Muhammad Sheraz	14	5	70	Figma screens, frontend UI, responsive pages, worker profile and recommendation cards.
Risk		Impact		Mitigation
Worker ID upload security		Sensitive document files may be mishandled.		Validate file type/size, store in protected media folder, restrict admin-only access.

Recommendation accuracy in early data	Low platform data may reduce quality of matching.	Use rule-based weighted scoring in Sprint 1 and improve with more data later.
UI consistency	Different screens may look inconsistent.	Use shared Figma components and common CSS/component library.
Scope creep	Too many AI features may overload Sprint 1.	Keep Sprint 1 AI to worker ranking only; move sentiment and pay suggestion to Sprint 2.

4.4.2 Sprint 1 Backlog

SP-ID	Task Name	Related PB-ID	Assigned To	Status	Est. Hrs	Actual Hrs	Comments
S1-01	Set up GitHub repository with main/development/feature branches	N/A	Ali Husnain	Complete	3	3	Repository created and branching strategy documented.
S1-02	Create Jira project, epics, user stories, and Sprint 1 board	N/A	Ali Husnain	Complete	4	4	Jira backlog structured and Sprint 1 started.
S1-03	Prepare base Figma flows for landing, registration, dashboards, job post, profile, and admin verification	UI-01	Muhammad Sheraz	Complete	8	8	Core screens prepared and reviewed.
S1-04	Design and implement user registration/login UI	AM-01	Muhammad Sheraz	Complete	8	7	Role-based UI completed.
S1-05	Implement authentication backend and role-based access	AM-01	Husnain Azam	Complete	10	10	Client, Worker, Admin roles configured.
S1-06	Build user profile update module	AM-02	Husnain Azam	Complete	6	6	Profile fields validated and saved.
S1-07	Build worker profile creation form	WP-01	Muhammad Sheraz	Complete	8	8	Skills, category, city, and availability captured.
S1-08	Implement national ID upload and storage validation	WP-02	Husnain Azam	Complete	8	9	File type and size validation added.
S1-09	Create admin verification queue UI	AD-01	Ali Husnain	Complete	7	7	Admin can view pending worker requests.
S1-10	Implement approve/reject verification workflow	AD-01	Husnain Azam	Complete	8	8	Worker status changes to Approved or Rejected.
S1-11	Create client job posting form and backend model	JP-01	Ali Husnain	Complete	10	10	Job category, city, area, duration, and pay rate stored.
S1-12	Develop worker job browse and apply feature	BK-01	Ali Husnain	Complete	8	8	Applications are stored as Pending.
S1-13	Implement initial smart recommendation scoring service	MT-01	Husnain Azam	Complete	12	13	Score formula implemented for verified workers.
S1-14	Integrate recommendation output with client job post success screen	MT-01	Muhammad Sheraz	Complete	6	6	Ranked worker cards displayed.
S1-15	Write Sprint 1 test cases and execute smoke testing	All S1	All Members	Complete	8	8	Core happy-path and validation tests passed.

Updated Product Backlog After Grooming

The backlog was groomed by keeping Sprint 1 focused on foundation and core marketplace flow, while Sprint 2 was planned for booking completion, trust feedback, AI enhancements and operational dashboards. The same backlog items should be reflected on the Jira board as Epics, Stories and Tasks.

PB-ID	Epic	User Story	Priority	Story Points	Sprint	Status	Acceptance Criteria
AM-01	Account Management	As a client/worker, I want to register and log in so that I can access role-based QuickHire features.	Must	8	Sprint 1	Done	Email must be unique; role must be selected; user is redirected to correct dashboard.
AM-02	Account Management	As a user, I want to manage my profile and contact information so that my account stays accurate.	Should	5	Sprint 1	Done	Profile changes save successfully and validation errors are shown.
WP-01	Worker Verification	As a worker, I want to create a verified profile with skills, city, experience, and	Must	8	Sprint 1	Done	Worker profile includes category, experience, area, availability, and profile status.

		availability so that clients can trust and hire me.					
WP-02	Worker Verification	As a worker, I want to upload my national ID document so that the admin can verify my identity.	Must	8	Sprint 1	Done	Only allowed file types are uploaded and worker status becomes Pending Verification.
AD-01	Admin Operations	As an admin, I want to approve or reject worker verification requests so that only verified workers appear publicly.	Must	8	Sprint 1	Done	Admin can view document, approve/reject, and worker receives status update.
JP-01	Job Posting	As a client, I want to post a short-term job with category, city, duration, and pay rate so that suitable workers can apply.	Must	8	Sprint 1	Done	Job is saved as Open and visible to relevant verified workers.
MT-01	AI Matching	As a client, I want to see AI-recommended workers after posting a job so that I can choose the best match quickly.	Must	13	Sprint 1	Done	Workers are ranked using category, location, rating, availability, and experience.
BK-01	Applications	As a worker, I want to browse and apply for matching jobs so that I can find short-term work.	Must	8	Sprint 1	Done	Application is stored as Pending and client receives notification.
BK-02	Booking Management	As a client, I want to confirm, reschedule, or cancel a booking so that I can manage job arrangements flexibly.	Must	13	Sprint 2	Planned	Booking status follows Pending, Confirmed, Completed, or Cancelled.
RR-01	Ratings & Reviews	As a client, I want to rate and review a worker after completion so that future clients can make informed decisions.	Must	8	Sprint 2	Planned	Review is accepted only after completed booking.
RR-02	Ratings & Reviews	As a worker, I want to rate my client after a job so that platform trust is maintained on both sides.	Should	5	Sprint 2	Planned	Worker review is attached to completed booking.
AI-02	Review Sentiment	As an admin/client, I want review sentiment to be classified so that feedback can be summarized clearly.	Should	8	Sprint 2	Planned	Review text is classified as Positive, Neutral, or Negative.
AI-03	Smart Pay Rate	As a client, I want a suggested pay rate while creating a job so that pricing is fair and competitive.	Should	8	Sprint 2	Planned	System displays suggested pay range based on category, city, and duration.
AD-02	Admin Dashboard	As an admin, I want dashboard analytics for users, bookings, and disputes so that I can monitor platform health.	Should	8	Sprint 2	Planned	Dashboard shows totals, recent bookings, pending verifications, and disputes.
NF-01	Quality	As a user, I want the platform to be responsive and secure so that I can use it safely on desktop and mobile.	Must	8	Sprint 2	Planned	Pages are responsive and sensitive actions use authentication/authorization.
NF-02	Notifications	As a user, I want email and in-app notifications for booking status changes so that I stay informed.	Should	5	Sprint 2	Planned	Notifications are triggered on application, verification, and booking status updates.

Jira Update Summary

Jira Artifact	Required Action	Status for Submission
Project Board	Create QuickHire Scrum board with Product Backlog, Sprint 1 and Sprint 2.	Backlog content prepared; import CSV included with this deliverable.
Epics	Create epics: Account Management, Worker Verification, Job Posting, AI Matching, Booking, Ratings & Reviews, Admin Operations, Quality.	Defined in Product Backlog table.
Sprint 1	Add selected Sprint 1 stories and mark all completed after Sprint 1 Review.	Sprint 1 tasks and statuses documented.
Sprint 2	Add planned Sprint 2 tasks, assignees and estimates.	Sprint 2 backlog prepared.
Evidence	Take screenshots of Jira backlog, active sprint, completed Sprint 1 and Sprint 2 board.	Insert screenshots in final FYP master document/presentation.

UI/UX Design Update and Figma Reference

The UI/UX design should be updated in Figma for the complete system flow. Sprint 1 requires the core screens needed for authentication, worker profile, worker verification, client job posting, job browsing and recommendation display. Sprint 2 requires booking lifecycle, review forms, dashboard widgets, and responsive refinements.

PID	Screen / Flow	Role	Sprint	Update Required
PID-01	Landing page with service categories and call-to-action	Visitor	Sprint 1	Confirm content, responsive hero and category cards.
PID-02	Registration and login screens with role selection	Client/Worker/Admin	Sprint 1	Complete. Add validation error states.
PID-03	Client dashboard	Client	Sprint 1-2	Add job status cards and booking summary.
PID-04	Worker dashboard/profile setup	Worker	Sprint 1	Complete profile, skill, city, experience and availability fields.
PID-05	National ID upload and verification status	Worker/Admin	Sprint 1	Add Pending, Approved, Rejected states.
PID-06	Create job post form	Client	Sprint 1	Add category, city, area, duration, pay rate and description parser placeholder.
PID-07	AI recommended worker cards	Client	Sprint 1	Show ranking score factors and worker summary.
PID-08	Worker job listing and application form	Worker	Sprint 1	Show open jobs and apply action.
PID-09	Booking management detail screen	Client/Worker	Sprint 2	Add confirm, cancel, complete and reschedule states.
PID-10	Two-way review form and sentiment result	Client/Worker/Admin	Sprint 2	Add rating, comment and sentiment tags.
PID-11	Admin dashboard analytics	Admin	Sprint 2	Add users, bookings, pending verifications and disputes widgets.

Figma link placeholder: Insert the actual QuickHire Figma prototype URL and screenshots before final submission if your supervisor requires visual evidence inside the document.

4.4.3 Sprint 1 Design Class Diagram

The Sprint 1 class diagram covers the selected backlog items: account management, worker profile and verification, client job post, worker application, booking foundation, review placeholder, recommendation engine and notification service.

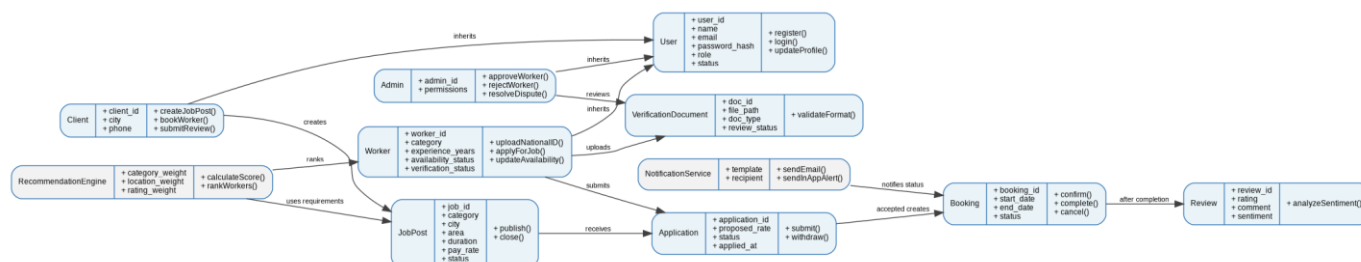


Figure 1: Sprint 1 Design Class Diagram - QuickHire Core Marketplace Classes

4.4.4 Sprint 1 Sequence Diagrams for Each User Story

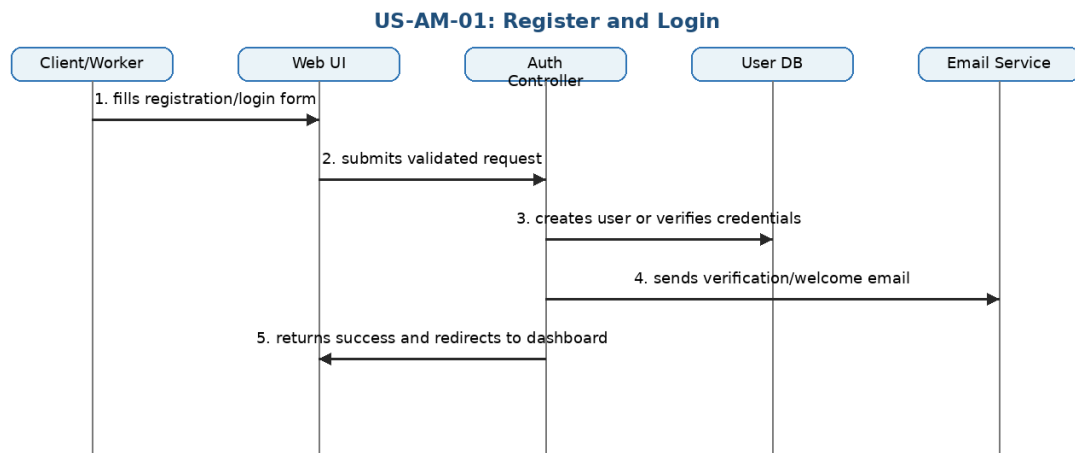


Figure 2: Sequence Diagram - Register and Login

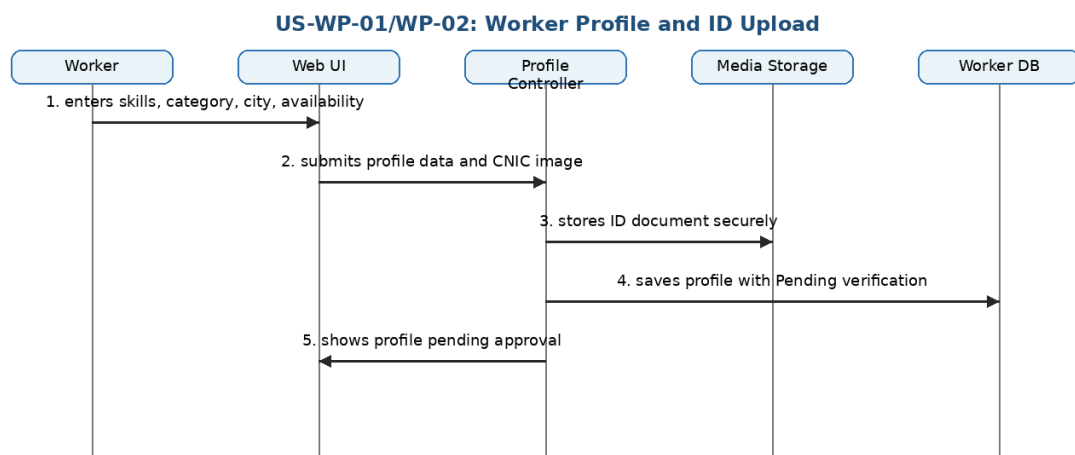


Figure 3: Sequence Diagram - Worker Profile and ID Upload

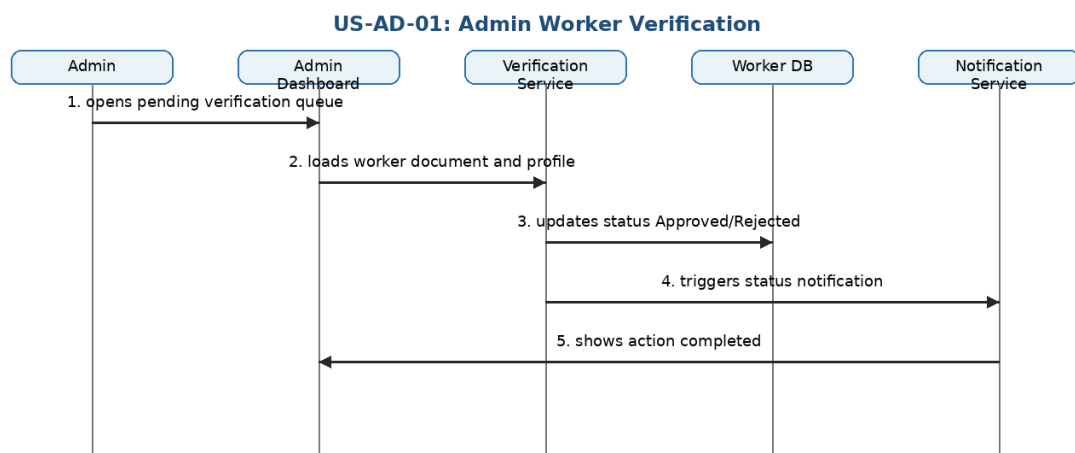


Figure 4: Sequence Diagram - Admin Worker Verification

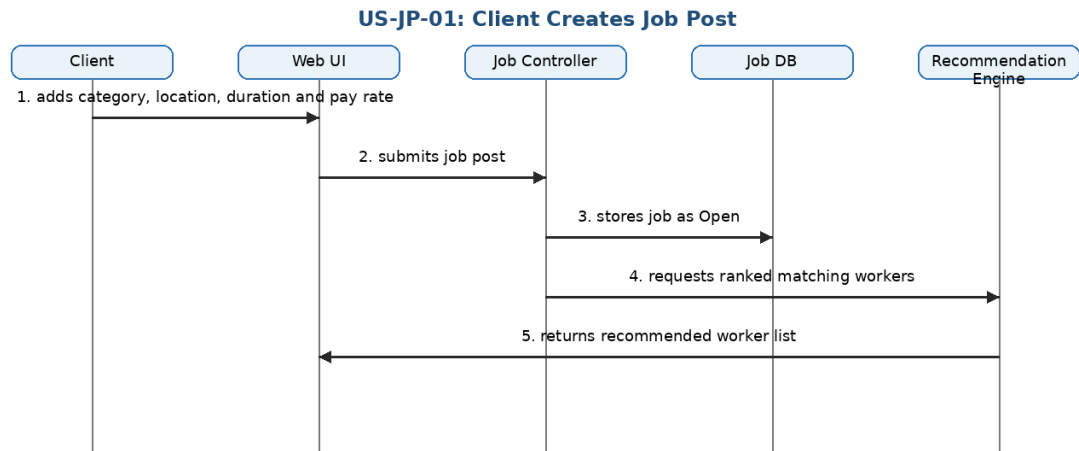


Figure 5: Sequence Diagram - Client Creates Job Post

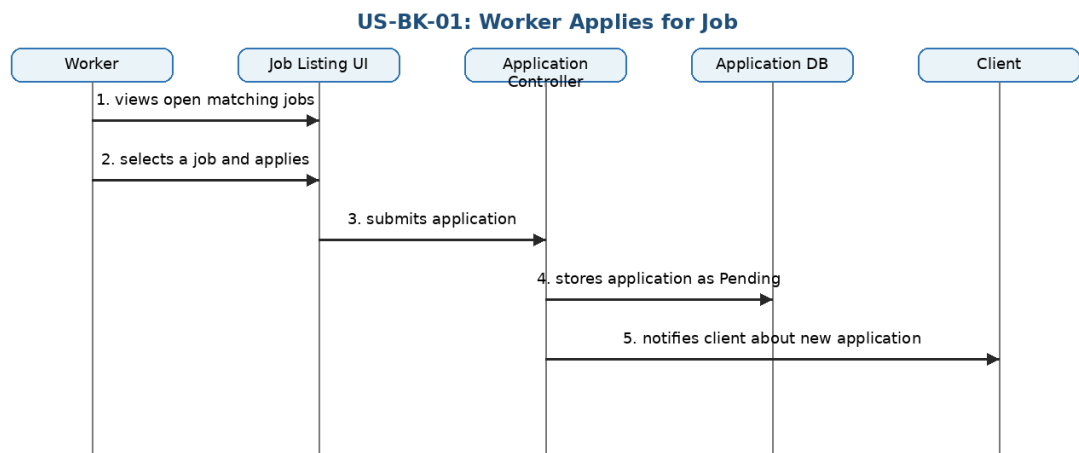


Figure 6: Sequence Diagram - Worker Applies for Job

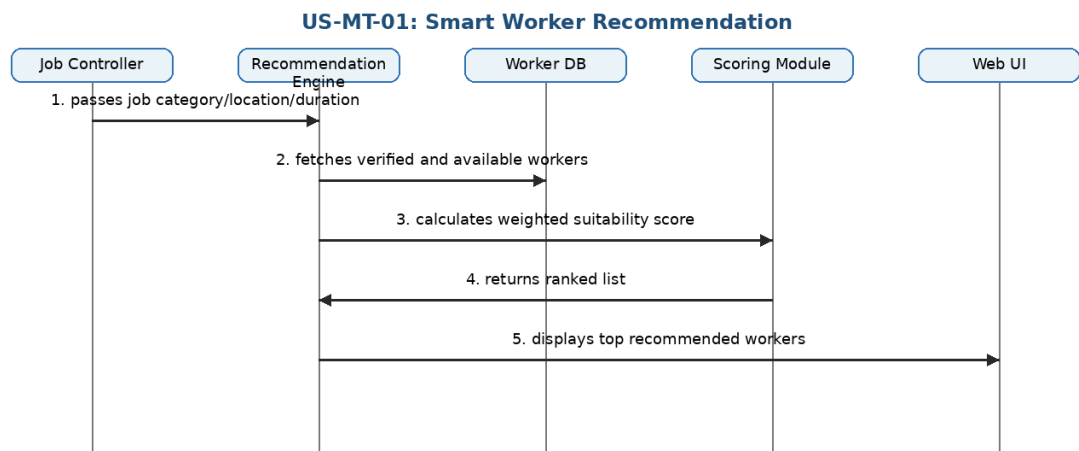


Figure 7: Sequence Diagram - Smart Worker Recommendation

4.4.5 Sprint 1 Decision Tables

Decision Table 1: Registration and Role-Based Access

Condition / Rule	R1	R2	R3	R4
Email is valid and unique	Y	Y	N	Y
Password meets policy	Y	N	Y	Y
Role selected as Client/Worker/Admin	Y	Y	Y	N
Action: Create account	Y	N	N	N
Action: Show validation error	N	Y	Y	Y
Action: Redirect to role dashboard	Y	N	N	N

Decision Table 2: Worker Verification

Condition / Rule	R1	R2	R3	R4
Worker profile complete	Y	Y	N	Y
National ID uploaded in valid format	Y	N	Y	Y
Admin approves document	Y	-	-	N
Action: Mark worker Approved	Y	N	N	N
Action: Keep Pending / Request correction	N	Y	Y	N
Action: Mark Rejected and notify worker	N	N	N	Y

Decision Table 3: Job Application Eligibility

Condition / Rule	R1	R2	R3	R4	R5
Worker account is active	Y	Y	Y	N	Y
Worker is verified	Y	N	Y	Y	Y
Job status is Open	Y	Y	N	Y	Y
Worker category/location matches job	Y	Y	Y	Y	N
Action: Allow application	Y	N	N	N	N
Action: Show reason and block application	N	Y	Y	Y	Y

4.4.6 Sprint 1 Extended Test Cases

TID-01 - Verify registration and login with valid Client data

Field			Value		
Test Case ID			TID-01		
Epic/Backlog ID			AM-01		
Test Priority			High		
Description			A new client can register, verify email and log in to the client dashboard.		
Pre-Condition			Required user account exists and relevant Sprint 1 module is deployed on test environment.		
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Open registration page	Role: Client	Registration form displayed	As expected	Pass
2	Enter valid name, email, phone and password	ali.client@test.com / Strong@12345	Fields accepted	As expected	Pass
3	Submit registration form	Valid form data	Account created and verification/welcome message shown	As expected	Pass
4	Log in with same credentials	Valid email/password	Client dashboard opens	As expected	Pass

Post Condition: Client account exists and user can access client dashboard.

TID-02 - Verify worker profile creation and national ID upload

Field			Value		
Test Case ID			TID-02		
Epic/Backlog ID			WP-01/WP-02		
Test Priority			High		
Description			A worker can complete a profile and upload a valid national ID document.		
Pre-Condition			Required user account exists and relevant Sprint 1 module is deployed on test environment.		
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Log in as worker	Verified login credentials	Worker dashboard displayed	As expected	Pass

2	Enter skills, category, experience, city and availability	Cook / Lahore / 3 years	Profile data accepted	As expected	Pass
3	Upload valid CNIC image/PDF	cnic_front.jpg	File is uploaded successfully	As expected	Pass
4	Save profile	Complete profile	Status becomes Pending Verification	As expected	Pass

Post Condition: Worker profile is saved and appears in admin verification queue.

TID-03 - Verify admin approval of worker verification request

Field			Value		
Test Case ID			TID-03		
Epic/Backlog ID			AD-01		
Test Priority			High		
Description			Admin can approve or reject a pending worker after document review.		
Pre-Condition			Required user account exists and relevant Sprint 1 module is deployed on test environment.		
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Log in as admin	Admin credentials	Admin dashboard displayed	As expected	Pass
2	Open pending verifications	Pending worker profile	Worker document and profile details displayed	As expected	Pass
3	Click Approve	Valid worker document	Worker status changes to Approved	As expected	Pass
4	Check worker public visibility	Approved worker	Worker appears in recommendation/search results	As expected	Pass

Post Condition: Worker is approved and eligible for job matching.

TID-04 - Verify client job post creation

Field			Value		
Test Case ID			TID-04		
Epic/Backlog ID			JP-01		
Test Priority			High		
Description			Client can publish a short-term job with all required fields.		
Pre-Condition			Required user account exists and relevant Sprint 1 module is deployed on test environment.		
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Open create job page	Client logged in	Job form displayed	As expected	Pass
2	Enter job details	Childcare / Lahore / 3 weeks / Rs. 2000 per day	Form validates required fields	As expected	Pass
3	Submit job	Valid job data	Job saved with Open status	As expected	Pass
4	View job listing	Open jobs list	New job appears for matching workers	As expected	Pass

Post Condition: Job is stored and visible to eligible workers.

TID-05 - Verify worker applies for a matching job

Field			Value		
Test Case ID			TID-05		
Epic/Backlog ID			BK-01		
Test Priority			High		
Description			Verified worker can apply for an open matching job.		
Pre-Condition			Required user account exists and relevant Sprint 1 module is deployed on test environment.		
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Log in as verified worker	Worker credentials	Worker dashboard displayed	As expected	Pass
2	Open matching jobs	Worker category/location	Relevant open jobs displayed	As expected	Pass
3	Click Apply	Selected job	Application form opens	As expected	Pass
4	Submit application	Optional note/proposed rate	Application saved as Pending	As expected	Pass

Post Condition: Client can see the worker application in job details.

TID-06 - Verify smart worker recommendation ranking

Field			Value		
Test Case ID			TID-06		
Epic/Backlog ID			MT-01		
Test Priority			High		
Description			System ranks workers after a client creates a job post.		
Pre-Condition			Required user account exists and relevant Sprint 1 module is deployed on test environment.		
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Create job post	Category: Cook, City: Lahore	Job saved	As expected	Pass
2	Trigger recommendation engine	Job post data	Verified and available workers fetched	As expected	Pass
3	Calculate worker score	Category, location, rating, availability, experience	Scores calculated correctly	As expected	Pass
4	Display ranked workers	Top scoring workers	Recommended workers shown in descending score order	As expected	Pass

Post Condition: Client receives a ranked list of matching workers.

4.4.7 Sprint 1 Review Meeting

Field		Details		
Sprint		Sprint 1 - Foundation and Core Marketplace Flow		
Review Date		Tuesday, 02 June 2026		
Duration		1 hour		
Facilitator		Dr. Ehtesham ul haq Dar / Ali Husnain		
Attendees		Ali Husnain, Husnain Azam, Muhammad Sheraz		
Sprint Goal Achievement		Achieved. Authentication, worker profile, ID upload, admin verification, job posting, worker application and initial recommendation flow were completed.		
ID	Task Name	Owner	Status	Review Notes
S1-01	Set up GitHub repository with main/development/feature branches	Ali Husnain	Complete	Repository created and branching strategy documented.
S1-02	Create Jira project, epics, user stories, and Sprint 1 board	Ali Husnain	Complete	Jira backlog structured and Sprint 1 started.
S1-03	Prepare base Figma flows for landing, registration, dashboards, job post, profile, and admin verification	Muhammad Sheraz	Complete	Core screens prepared and reviewed.
S1-04	Design and implement user registration/login UI	Muhammad Sheraz	Complete	Role-based UI completed.
S1-05	Implement authentication backend and role-based access	Husnain Azam	Complete	Client, Worker, Admin roles configured.
S1-06	Build user profile update module	Husnain Azam	Complete	Profile fields validated and saved.
S1-07	Build worker profile creation form	Muhammad Sheraz	Complete	Skills, category, city, and availability captured.
S1-08	Implement national ID upload and storage validation	Husnain Azam	Complete	File type and size validation added.
S1-09	Create admin verification queue UI	Ali Husnain	Complete	Admin can view pending worker requests.
S1-10	Implement approve/reject verification workflow	Husnain Azam	Complete	Worker status changes to Approved or Rejected.
S1-11	Create client job posting form and backend model	Ali Husnain	Complete	Job category, city, area, duration, and pay rate stored.
S1-12	Develop worker job browse and apply feature	Ali Husnain	Complete	Applications are stored as Pending.
S1-13	Implement initial smart recommendation scoring service	Husnain Azam	Complete	Score formula implemented for verified workers.
S1-14	Integrate recommendation output with client job post success screen	Muhammad Sheraz	Complete	Ranked worker cards displayed.
S1-15	Write Sprint 1 test cases and execute smoke testing	All Members	Complete	Core happy-path and validation tests passed.
Demo Area		Outcome		
Authentication		Client, Worker and Admin users were able to log in and access role-based dashboards.		
Worker Verification		Worker profile and national ID upload completed; admin approval workflow demonstrated.		
Client Job Posting		Client created a short-term job and saved it as Open.		
Worker Application		Verified worker viewed relevant open jobs and applied successfully.		
Smart Recommendation		Initial score-based ranking displayed matching workers after job posting.		
Metric		Value		
Committed Story Points		69		
Completed Story Points		69		
Carryover Items		0		

Bugs Reported / Resolved	4 / 4	
Sprint Goal Status	Completed	
Feedback Source	Comments	Action
Supervisor	Scope is relevant; maintain clear documentation and tool evidence.	Update Chapter 4 artifacts and include required meeting minutes.
Internal QA	File upload validation and role redirects must be tested carefully.	Added validation test cases and smoke tests.
Team	Recommendation formula should remain simple in Sprint 1.	Move sentiment and pay suggestion to Sprint 2.

4.4.8 Sprint 1 Retrospective Meeting

Field	Details	
Retrospective Date	Tuesday, 02 June 2026	
Duration	45 minutes	
Facilitator	Ali Husnain	
Attendees	Ali Husnain, Husnain Azam, Muhammad Sheraz	
Sprint Overview	Sprint 1 delivered the foundation and first marketplace flow from user registration to job application and recommendations.	
What Went Well	Evidence	
Core marketplace workflow was completed within sprint.	All Sprint 1 backlog tasks marked Complete.	
Team used Jira-style task breakdown and GitHub branching.	Backlog IDs and task ownership are traceable.	
Figma helped clarify screens before implementation.	Sprint 1 UI screens mapped to prototype IDs.	
AI recommendation was implemented as a controlled score-based service.	Scoring factors match project objectives.	
What Did Not Go Well	Improvement	
Some UI validation states were identified late.	Define validation states in Figma before development.	
File upload security took extra time.	Prepare reusable secure upload helper in Sprint 2.	
Tool evidence screenshots were not captured daily.	Capture Jira, Slack, GitHub and Figma screenshots at the end of each meeting/day.	
Action Item	Owner	Due Date
Add responsive refinements for dashboard and job screens	Muhammad Sheraz	Sprint 2 Week 4
Improve API documentation and add more backend validation	Husnain Azam	Sprint 2 Week 2
Maintain Jira status updates and prepare Sprint 2 evidence screenshots	Ali Husnain	Ongoing
Expand tests for booking, reviews and notifications	All Members	Sprint 2 Week 4

4.5 Sprint 2

4.5.1 Sprint 2 Planning Meeting Minutes

Field	Details	
Meeting Title	Sprint 2 Planning Meeting - Booking, Reviews, AI Enhancements and Admin Analytics	
Date	Wednesday, 03 June 2026	
Time / Duration	10:00 AM - 11:20 AM / 80 minutes	
Facilitator	Dr. Ehtesham ul haq Dar / Scrum Master: Ali Husnain	
Attendees	Ali Husnain, Husnain Azam, Muhammad Sheraz	
Sprint Duration	03 June 2026 - 24 June 2026	
Sprint Goal	Complete the booking lifecycle and trust feedback loop by adding booking status management, two-way reviews, sentiment analysis, smart pay rate suggestion, notifications, admin analytics and responsive UI improvements.	
Selected Backlog Item	Reason	Assigned Owner
BK-02 Booking Management	Required to move from application to confirmed work arrangement.	Ali Husnain / Husnain Azam
RR-01 Client Review	Creates trust for future clients.	Muhammad Sheraz / Husnain Azam
RR-02 Worker Review	Completes two-way accountability.	Muhammad Sheraz / Husnain Azam
AI-02 Sentiment Analysis	Summarizes review quality and supports moderation.	Husnain Azam
AI-03 Smart Pay Rate Suggestion	Reduces pricing uncertainty during job posting.	Ali Husnain

AD-02 Admin Dashboard	Provides operational visibility for the platform.	Ali Husnain
NF-02 Notifications	Keeps users informed about booking and verification changes.	Ali Husnain
NF-01 Responsive UI Updates	Improves quality of system demo and usability.	Muhammad Sheraz
Risk	Impact	Mitigation
Sentiment model accuracy may be limited with small review dataset.	Incorrect sentiment tags may affect profile trust.	Use VADER/TextBlob initially and allow admin moderation.
Booking status transitions may become inconsistent.	Users may see wrong booking state.	Use clear state machine: Pending, Confirmed, Completed, Cancelled.
Notification failures may confuse users.	Users may miss important status updates.	Add in-app notifications as fallback and log email status.
Sprint 2 scope is larger than Sprint 1.	Incomplete features may carry over.	Prioritize booking and reviews first; analytics and refinements second.
Member	Sprint 2 Capacity	Primary Tasks
Ali Husnain	75 hours	Booking dashboard integration, pay rate suggestion, admin analytics, notifications, Jira/Slack evidence.
Husnain Azam	75 hours	Booking APIs, review/rating backend, sentiment analysis, data validation and testing.
Muhammad Sheraz	70 hours	Booking/review UI, responsive refinements, Figma updates and UI test support.

4.5.2 Sprint 2 Backlog

SP-ID	Task Name	Related PB-ID	Assigned To	Status	Est. Hrs	Priority	Timeline
S2-01	Design booking detail and status update screens	BK-02	Muhammad Sheraz	To Do	6	High	Sprint 2 Week 1
S2-02	Implement booking confirmation, cancellation, and completion APIs	BK-02	Husnain Azam	To Do	12	High	Sprint 2 Week 1-2
S2-03	Connect booking status changes with client and worker dashboards	BK-02	Ali Husnain	To Do	8	High	Sprint 2 Week 2
S2-04	Build client-to-worker review form	RR-01	Muhammad Sheraz	To Do	6	High	Sprint 2 Week 2
S2-05	Build worker-to-client review form	RR-02	Muhammad Sheraz	To Do	5	Medium	Sprint 2 Week 2
S2-06	Implement review storage, rating aggregation, and validation rules	RR-01/RR-02	Husnain Azam	To Do	10	High	Sprint 2 Week 2-3
S2-07	Implement sentiment analysis using VADER/TextBlob	AI-02	Husnain Azam	To Do	8	Medium	Sprint 2 Week 3
S2-08	Implement smart pay rate suggestion service	AI-03	Ali Husnain	To Do	8	Medium	Sprint 2 Week 3
S2-09	Build admin dashboard analytics widgets	AD-02	Ali Husnain	To Do	8	Medium	Sprint 2 Week 3-4
S2-10	Implement email/in-app notification templates	NF-02	Ali Husnain	To Do	5	Medium	Sprint 2 Week 4
S2-11	Perform responsive UI updates from Figma to implementation	NF-01	Muhammad Sheraz	To Do	7	High	Sprint 2 Week 4
S2-12	Run Sprint 2 regression, test documentation, and bug fixing	All S2	All Members	To Do	10	High	Sprint 2 Week 4

Mandatory Tool Usage Evidence Checklist

Tool	Purpose	Evidence to Include in Final Presentation	Status
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Jira	Project backlog and task tracking	Backlog screenshot, Sprint 1 completed board, Sprint 2 active board, burndown if available.	Content prepared; add screenshots from actual account.
Slack	Team communication instead of WhatsApp	Channel screenshot showing sprint updates, blockers and meeting coordination.	Add screenshots from actual workspace.
GitHub	Code repository and version control	Repository URL, branches, commits, pull requests and issue references.	Add actual repository URL and screenshots.
GitHub Copilot	AI-assisted coding	Screenshot or explanation of Copilot usage in IDE for code suggestions/tests.	Add IDE screenshot if required.
Figma	UI/UX design	Prototype link and screen screenshots for Sprint 1/Sprint 2 flows.	Screen list prepared; add actual link/screenshots.